Relative Value Beyond the Pair Trade

Higher-Order Pricing Relationships, Their Enforcement Mechanisms, and Their Measurement

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Abstract. Relative value is usually studied as a property of a pair of prices. This paper studies it as a property of the mechanism that enforces the relationship between prices. We catalogue sixteen higher-order pricing relationships across rates, credit, repo, securitized products and market plumbing, and we classify each one by its enforcement mechanism: accounting identity, contract and statute, physical conversion, no-arbitrage subject to a balance-sheet band, and statistical regularity. For each relationship we derive the restriction, name the measurable variables, and report what a newly assembled record of public and regulatory data can measure. Three results stand out. First, clearing netting ratios are slow-moving state variables, not constants: over January 2001 to September 2026 the settled share of compared volume at the Government Securities Division averages 35.0 percent (standard deviation 4.7 percentage points), while the share of Mortgage-Backed Securities Division clearing par that settles on the Electronic Pool Notification system fell from 19.2 percent in 2001 to 2.6 percent in September 2026. Second, swap spreads implied by the September 2026 swap tape are negative at every tenor we can measure (medians of daily spreads: 2-year -13 bp, 10-year -39 bp, 30-year -70 bp), and a calibrated balance-sheet calculation puts the break-even 10-year spread near -30 bp, so the measured spreads sit inside the band that balance-sheet costs predict. Third, the largest apparent bases in our data are composition artefacts: a naively computed index-versus-basket gap for CDX North America Investment Grade averages -33 bp (standard deviation 14 bp) over September 2026 and contains no arbitrage. We close with three pre-specified tests for the highest-ranked relationships, each with a stated falsifier, and a list of relationships the public record cannot yet measure. Two of those tests have since been run on their stated terms, and both claims are withdrawn: the loan-tranche-ratings triangle (the collateral block added 0.0092 AUC on its holdout, below the pre-specified 0.01 threshold) and the balance-sheet factor test for the swap spread and the cross-currency basis. The third test, on the netting ratio as a state variable for settlement stress, proved not executable as designed, because the fails data its falsifier requires do not exist for its evaluation window; its claim is neither supported nor withdrawn.

Keywords: relative value; limits to arbitrage; balance sheet costs; clearing and netting; covered interest parity; swap spreads; convenience yield; CDS-bond basis; credit ratings; market microstructure

JEL codes: G12, G13, G14, E44.

Non-technical summary. Two prices that should be equal usually are not, and the gap is usually not free money. This paper asks, for sixteen of the best-known gaps in modern markets, three questions. What force is supposed to close the gap? How large is the gap in the data we can actually obtain? And when the gap is large, is it a profit opportunity, a payment for some risk or cost, or an error in how the gap was measured? The answers differ sharply by relationship. Some gaps are closed by contract or by accounting identity and can be audited rather than traded. Some are closed by arbitrageurs who must rent bank balance sheet to trade, so the gap settles at whatever level pays the rent, and it widens when balance sheet is scarce. Some are not closed by any identifiable force and persist because they are prices of risk, not mistakes. And some gaps that look large and stable in the data are artefacts of comparing the wrong baskets. The practical conclusion is that a trader, a risk

manager or a regulator should ask what enforces a relationship before asking how large it is, because the enforcement mechanism, not the size of the gap, decides what happens next.

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1 Introduction

Most relative-value research starts from a pair of prices and asks whether their difference is too large. We start from the opposite end. Every candidate relationship in this paper is required to name, before any data is consulted, the mechanism that is supposed to enforce it: the contract, plant, clearing house, tax schedule or balance sheet whose operation pushes the prices back into line, and the loss state that a trader bears while waiting. Relationships that cannot name an enforcer are excluded from the catalogue regardless of how well they fit past data. Sixteen relationships pass that screen. They span clearing and settlement plumbing, interest rate swaps, cross-currency funding, Treasury convenience, repo specialness, the cash-futures basis, the CDS-bond basis, index construction, equity correlation, physical conversion margins, mortgage dollar rolls, perpetual futures, market segmentation, municipal taxation, securitized credit, and the marks that fund accountants place on illiquid bonds.

The screen is not a formality. It changes what counts as evidence. A gap enforced by an accounting identity cannot be traded, but it can be audited, and its deviations are measurement errors that locate broken data. A gap enforced by arbitrageurs who must rent balance sheet has an equilibrium width set by the rent, so its persistence is information about the price of balance sheet, not about market inefficiency. A gap with no enforcer is a risk premium wearing the clothes of a mispricing, and its backtest will look best exactly before the loss state arrives. Sorting the sixteen relationships into five enforcement classes (identity; contract and statute; physical conversion; no-arbitrage with a balance-sheet band; statistical regularity) is the paper's first task, and most of its conclusions are corollaries of the sort.

The second task is measurement. We assembled the measurement base from public and regulatory sources rather than vendor feeds: the full public swap dissemination tape for September 2026 (60,245,115 trades (Observed, A)), FICC clearing volumes published monthly since January 2001, daily repo and settlement-fails files, the complete ABS-EE and Form 10-D record for registered CMBS, 28.3 million rating-action events, fund holdings from Form N-PORT, and a corporate bond print record of 4.1 million trades. Section 4 documents each source. The base is deep in some directions and thin in others, and we report both. Several relationships in the catalogue cannot be measured on it at all; we say which, and why, rather than reach for proxies.

Three measured results preview the argument.

First, netting is a state variable. The netting ratio of the Government Securities Division (GSD) of FICC, settled volume divided by compared volume, averages 35.0 percent over 309 months with a standard deviation of 4.7 percentage points (Observed, A); its multiplier, the gross exposure extinguished per unit of settled exposure, moved from 3.37 in January 2001 to 3.06 in September 2026 while gross volume grew 8.8-fold. In the Mortgage-Backed Securities Division (MBSD), the share of clearing par settling through the Electronic Pool Notification (EPN) system fell from 19.2 percent

(2001 average) to 2.6 percent in September 2026 (OBSERVED, A), a twenty-five-year migration of settlement into netting. Any model that treats netting efficiency as a constant misstates the funding and margin consequences of cleared volume by a first-order amount.

Second, the negative swap spread is an equilibrium price, and we can measure it on the trade tape. Swap spreads implied by median fixed rates on new trades in September 2026, against same-day Treasury par yields, are −13 bp at two years, −28 bp at five years, −39 bp at ten years and −70 bp at thirty years (medians of daily spreads) (RECONSTRUCTED, B). A deliberately simple break-even calculation, set out in Section 7, puts the balance-sheet break-even for the ten-year point near −30 bp. The measured spread is more negative than the single-dealer break-even, which is the expected configuration when the marginal holder of the trade is a leveraged non-dealer rather than a dealer, and it is far from the region that a mispricing reading would require. The parallel cross-currency basis is the same phenomenon in a second market, which is why we treat the two as one relationship with two measurement sites and design a joint factor test for them (Appendix B).

Third, the largest apparent basis in our data is an artefact. Computing a CDS index-versus-constituent gap on the tape the naive way (the CDX North America Investment Grade index against the median of a 184-name segment basket) yields a gap averaging −33 bp with a standard deviation of 14 bp over the 21 trading days of September 2026 (RECONSTRUCTED, B). The number is stable, statistically tight, and empty: the basket is not the index constituent set, and the gap is composition. We keep this exhibit in the paper because it is the cleanest demonstration of the failure mode the catalogue is built to avoid, and because the same construction, applied to the correct constituents, is a genuine additivity restriction that the tape can support.

The paper also reaches negative conclusions, which we state with the same prominence. The equity correlation risk premium is genuine in the literature and unmeasurable here, because no listed option prices are held and the equity tape does not identify underliers. TBA dollar-roll specialness cannot be computed, because our mortgage data are non-agency. Perpetual futures funding cannot be computed, because we hold no crypto market data. The municipal ratio cannot be computed, because we hold no municipal panel. Each of these sections is written in full (mechanism, measurement design, breakdown conditions) with the data status stated in its first paragraphs, because the framework is the deliverable for those relationships and the empirics are deferred to data we do not hold.

Two relationships are ranked highest for immediate work, and Appendix B gives both a test design with a falsifier stated in advance. The loan-tranche-ratings triangle in registered CMBS (Section 20) is the only relationship in the catalogue where all three legs are observable at scale in our data: the collateral tape, the tranche stack it determines, and the rating actions that respond to it, with the response made observable by Rule 17g-7 disclosure and made urgent by its expiry (agency files end in October and November 2025, leaving roughly a year of collateral history with no observable rating response). The netting identity (Section 6) ranks second: it is the relationship on which all cleared-market risk arithmetic silently depends, and its ratio component, as distinct from its volume component, is a candidate state variable for settlement stress. The falsifiability status of the paper as a whole is stated in Section 23: the catalogue and measurements are retrospective, with no pre-registered analysis plan, except for the three test designs in Appendix B, which were stated before their holdout samples were touched. All three tests have since been taken to their stated endpoints: Test 1 was run on its holdout and its claim was withdrawn; Test 2 proved not executable as designed, because the fails data its falsifier requires do not exist for the evaluation window, so its claim is neither supported nor withdrawn; Test 3 was run and its claim was withdrawn.

The rest of the paper proceeds as follows. Section 2 sets out the institutional background. Section 3 reviews the related literature. Section 4 describes the data, with summary statistics. Section 5 presents the enforcement-class framework and the state-space form we use throughout. Sections 6 to

21 take the sixteen relationships in turn. Section 22 treats persistence, regime shifts and costs across the catalogue. Section 23 states limitations and robustness, and Section 24 concludes. Appendix A lists the ideas we examined and rejected, with reasons. Appendix B states the three pre-specified tests and reports the outcome of each. Appendix C is the data appendix.

2 Institutional background

The relationships in the catalogue live inside a specific plumbing system. This section records the parts of that system the later sections rely on, at the level of detail the measurements require.

2.1 Clearing at FICC

The Fixed Income Clearing Corporation (FICC) clears United States government securities through its Government Securities Division (GSD) and agency mortgage-backed securities through its Mortgage-Backed Securities Division (MBSD). In GSD, a trade is *compared* when both sides' submissions match; comparison creates the gross population of obligations. Netting then compresses each participant's compared obligations, by security and settlement date, into a single net delivery or receipt obligation; only the net is *settled*. The ratio of settled to compared volume, and the settlement channels through which MBSD obligations are discharged, are published monthly and are the raw material of Section 6. Within MBSD, pools can be settled through the Electronic Pool Notification (EPN) system, under which specific pools are allocated against TBA obligations, or through netting processes that extinguish obligations without pool delivery. The EPN share of clearing par is therefore a direct read on how much of the TBA market still settles by allocation.

FICC also operates the GCF Repo service, a centrally cleared general-collateral repo market whose daily index (par amounts and weighted average rates for Treasury and MBS collateral) is the aggregate leg of the specialness analysis in Section 10, and the Sponsored Service, under which a sponsoring member clears repo for sponsored members. Sponsored volumes are the fastest-growing part of the cleared complex: the daily total of repo and reverse repo rose from a 2021 average of $0.16 trillion to $2.44 trillion in 2026 (through 15 October) (Observed, A). Settlement fails are published daily by security class; Treasury fails averaged $30.9 billion per day over the year to 20 October 2026 (Observed, A), against GSD compared volume of roughly $250 trillion per month, so fails are a thin tail of the cleared flow in normal times and a diagnostic in stressed ones.

2.2 The swap dissemination tape

Since the Dodd-Frank reforms, swaps within scope are reported to swap data repositories and publicly disseminated with capped notional disclosure. The DTCC public tape is disseminated in two jurisdictions (CFTC and SEC security-based swaps) and five asset classes. The September 2026 partitions, which we parsed in full, contain 60,245,115 trades (Observed, A). Three features of the tape shape everything we can measure from it. First, the dissemination *price* field is unpopulated for rates, credit and FX trades; economic prices live in class-specific fields (the fixed rate on a rates trade, the spread on a credit trade, the forward points on an FX trade), so every price in this paper is reconstructed from those fields. Second, notional caps bind on a large share of trades (66.2 percent of SEC credit trades in September 2026 (Observed, A)), so tape notionals are floors, not quantities. Third, the tape is a record of new trades and lifecycle events, not of positions; exposures must be accumulated, and the accumulation is exact only within the parsed window.

2.3 Structured credit disclosure

Registered securitizations disclose at three levels. Asset-level data under Regulation AB II (Form ABS-EE) give the loan tape: for registered CMBS, 28,638 loans across 517 deals in our build (Observed, A). Monthly distribution reports on Form 10-D give the tranche side: balances, factors and credit support as reported by certificate administrators (23,464 of 23,636 inventoried filings parsed cleanly in our build (Observed, A)). Rating actions are disclosed by the agencies under Rule 17g-7, in machine-readable form; our event layer holds 28,292,725 deduplicated events (Observed, A). The joint availability of the three legs, loan by loan and month by month, is what makes the triangle of Section 20 testable, and its limit is equally specific: the major agencies' 17g-7 files end in October and November 2025, so the rating leg is unobservable for the most recent year of collateral history.

2.4 Fund holdings and marks

Form N-PORT filings give monthly fund holdings with manager-assigned fair values. For the 707 corporate bonds in our monitoring universe, 550 (78 percent) carry at least one fund mark in the latest report period and 603 in at least one quarter (Observed, A). The site mark used in Section 21 is the cross-fund median, labelled by the fund report date; an earlier construction that labelled marks by the SEC report-ending-period field placed some marks in the future and was corrected on 20 October 2026 (5,699 bond labels changed; the values themselves were unaffected). Fund marks are accounting quantities produced under fair-value rules, and Section 21 treats them as such: their comparison with transaction prints is a study of two measurement systems, not of two prices for the same object.

2.5 The fails charge and the cost of not delivering

Since May 2009, the Treasury Market Practices Group fails charge has imposed a 3 percent per annum charge on failed Treasury deliveries (Treasury Market Practices Group, 2009). The charge converts a settlement fail from a free option into a priced one, and it caps repo specialness: a security cannot trade special at an implied rate more negative than roughly minus 3 percent for long, because failing becomes the cheaper way to be short. This single institutional fact disciplines the interpretation of both the fails data (Section 10) and the cash-futures basis (Section 11), and it is a clean example of the paper's organizing claim: the width of a relative-value band is set by an institutional parameter that can be read off a rulebook, not estimated from prices.

3 Related literature

The catalogue sits at the intersection of four literatures. We take them in the order the framework needs them.

Limits to arbitrage and balance-sheet costs. The modern reading of persistent price gaps is that arbitrage capital is intermediated and priced. For swap spreads, Aquilina et al. (2024) argue that negative spreads are the signature of pressure in government debt absorption: intermediaries must be compensated for holding bonds and paying fixed. Boyarchenko et al. (2018) compute the balance-sheet arithmetic of the arbitrage trade under the supplementary leverage ratio and show that the spread must be substantially negative before the trade clears a dealer's return hurdle; Adrian et al. (2017) link dealer balance-sheet capacity to bond market liquidity provision directly. Hanson et al. (2022) give the complementary demand-side account, in which the imbalance of duration supply

and hedging demand is itself a priced risk. We use all three as the costing layer of Sections 7 and 8: the band is computed from institutional charges, and the literature's role is to tell us which charges belong in the computation.

Covered interest parity and the dollar. Du et al. (2018) document that post-crisis CIP deviations are large, persistent and cross-sectionally systematic, and they attribute them to intermediary balance-sheet constraints rather than credit risk, using same-issuer bonds across currencies to strip issuer credit out of the calculation. Avdjiev et al. (2019) show the deviations load on a single dollar factor. Cenedese et al. (2019) document the calendar structure (quarter-end and year-end spikes) that ties the deviations to reporting dates and G-SIB surcharges. Our Section 8 adds no new CIP estimate; it specifies how the deviation is computed from the FX leg of the tape, why it belongs in the same factor as the swap spread, and what observation would falsify the unification.

Treasury convenience and collateral. Krishnamurthy and Vissing-Jorgensen (2012) estimate the Treasury convenience yield and its dependence on the supply of debt; the hedging-perspective manuscript from NYU Stern (NYU Stern School of Business, n.d.) ties convenience to the covariance of Treasury returns with equities. On the collateral side, Duffie (1996) derives the capitalization of specialness into bond prices; Cahill et al. (2013) and Cahill et al. (2018) measure the supply effects and cross-sectional pricing of special repo rates; the ECB study (Ampudia et al., 2025) estimates the effect of Eurosystem purchases on Bund specialness per unit of free float. Singh (2012) supplies the velocity arithmetic for pledged collateral that Section 5 uses to connect netting multipliers to the wider collateral system. Barth and Kahn (2021) document the scale of the Treasury cash-futures basis trade and its March 2020 unwind pattern, which Section 11 uses as the breakdown template.

Credit, ratings and demand systems. For the CDS-bond basis, Bai and Collin-Dufresne (2011) decompose crisis-period bases into funding, counterparty and liquidity components, and Augustin and Schnitzler (2020) show that funding illiquidity reinforces asset-specific illiquidity in the cross-section. For index products, the base-correlation construction and its skew have been documented since Bank for International Settlements (2005); for correlation as a priced risk, Driessen et al. (2009) estimate the premium and Bossu (2014) give the dispersion-trade mechanics. For ratings, Griffin and Tang (2012) is the template study of model-versus-agency gaps in structured credit; our Section 20 follows its identification logic (a fundamentals model, an agency outcome, a gap with a sign) on the post-crisis registered CMBS universe. For dollar rolls, Kitsul and Ochoa (2016) derive the funding-liquidity factor from roll implied financing rates and Song and Zhu (2019) characterize roll specialness. For municipal pricing, Wang et al. (2008) estimate the liquidity component of muni yields and Longstaff (2011) identifies the marginal tax rate and its premium in tax-exempt rates. For perpetual futures, Ackerer et al. (2024) derive the no-arbitrage price under general funding specifications and Angeris et al. (2022) give the mechanism primer. For holder-based estimation, Koijen and Yogo (2019) is the demand-system framework Section 5.4 adapts. None of these literatures is summarized further here; each relationship section engages the two or three papers that bind on it.

4 Data

We use eight data blocks, all drawn from public or regulatory sources and assembled into a local vault with standing update jobs. Appendix C lists every series with its source, retrieval date, coverage and known limitation; this section describes the blocks and reports summary statistics for the series the empirics use.

The swap tape. The DTCC public dissemination tape, CFTC and SEC partitions, September 2026 parsed in full: 60,245,115 trades (OBSERVED, A). The credit partitions hold 129,579 trades in the

month, from which we build the single-name panel: 1,251 reference entities, 193 with at least one priced five-year trade, 127 with at least three, on which all spread statistics in Sections 12 and 13 are computed. Monthly slices for July, August and October 2026 (the last through 19 October) are built by the same code; September is the reference month because it is the only complete month at full fidelity. Field discipline matters: the dissemination price field is populated on 0.0 percent of rates, credit and FX trades, prices being recovered from class-specific spread and rate fields; 66.2 percent of SEC credit trades carry capped notionals, so notionals are floors; the mandatory-clearing flag is set on 43.8 percent of CFTC rates flow and the package flag on 56.9 percent (Observed, A).

FICC publications. One file carries both clearing divisions' monthly volumes from January 2001 to September 2026 (309 months, 1,236 values) (DTCC, 2026b). The file's unit convention changes across its history: most values are recorded in decimal trillions, while the 2025 and 2026 values are recorded as full dollars with thousands separators. We normalize value by value using magnitude (any figure above 10^6 in the raw text is dollars), and we verify the normalization against the recorded anchors (January 2001 GSD compared $28.25 trillion; September 2026 $248.65 trillion). Ratios within a month are unit-free and unaffected. Daily files give the GCF repo index (250 trading days, 16 October 2025 to 16 October 2026), Treasury and agency fails (250 days, 20 October 2025 to 20 October 2026), and Sponsored Service volumes (1,289 days, 16 August 2021 to 15 October 2026) (DTCC, 2026a). The two 250-day files are rolling windows maintained by the source; their depth grows only from our first collection date, a limitation stated wherever they are used. The underwriting file gives monthly counts and amounts for four products (municipal, corporate, equity, certificates of deposit) from January 2011 to August 2026, 188 months (DTCC, 2026a).

Structured credit. The ABS-EE build covers registered offerings only: 46,595 filings from 776 issuers, 517 deal units, 28,638 loans and 38,855 property and component records. The 10-D build parsed 23,464 of 23,636 inventoried distribution reports, producing 738,118 tranche-month balance rows and a current-support table of 701,016 tranche-months, of which 94.3 percent carry a derived credit support figure (Observed, A). The ratings layer holds 28,292,725 deduplicated Rule 17g-7 events; the loan-deal-tranche join key is the tranche CUSIP, populated for 99.06 percent of tranches (10,999 of 11,103), and 76.4 percent of CUSIP classes join to at least one agency's current rating (Observed, A).

Holdings and prints. Form N-PORT gives the fund-mark panel (28 quarters; 612.6 million parsed rows across all tables in the full pull). The TRACE-based print store holds 4,100,179 trades as of the 21 October 2026 collection cycle (Observed, A), across a monitored universe of 707 bonds; the store is append-only.

Macro and money-market series. Treasury par yields (DGS2, DGS5, DGS10, DGS30), the ten-year breakeven inflation rate (T10YIE) and WTI crude (DCOILWTICO) come from FRED (Federal Reserve Bank of St. Louis, 2026), pulled on 22 October 2026 and verified observation by observation for the values quoted (WTI $91.48 on 1 September 2026 and $107.89 on 20 October 2026). Library series from the ElkassabgiData Econ Data Library (321 indexed sources) are used for context only (euro-area AAA curve, BIS effective exchange rates, EIA series), each cited with its producer and its *data_through* date; material figures are never taken from the library alone. One FRED series we pulled, the ICE BofA swap-rate series ICSAADR10Y, returned values inconsistent with any market reading (7.83 for the week of 25 September 2026, implying a swap spread near +270 bp against the same week's Treasury yield); we treat it as a series-definition mismatch, flag it here, and do not use it anywhere. Tape-implied rates are used instead.

Table 1: Summary statistics for the principal measured series

Series	N	Mean	SD	Min	Max
GSD netting ratio ν (%, monthly)	309	35.0	4.7	24.6	45.3
MBSD EPN share of clearing par (%, monthly)	309	9.8	4.7	2.6	25.8
GCF MBS less Treasury rate gap (bp, daily)	250	1.8	1.8	−3.5	10.2
Treasury fails ($bn, daily)	250	30.9	13.0	14.7	116.0
Single-name 5Y CDS spread (bp, Sep 2026 members)	127	78	75	13	598
Percentiles 10 / 25 / 50 / 75 / 90		29 / 43 / 59 / 88 / 130			
Tape swap spread, 2Y (bp, daily medians, Sep 2026)	21	−13.3	2.8	−20.6	−6.9
Tape swap spread, 5Y (bp)	21	−28.1	2.7	−34.7	−23.2
Tape swap spread, 10Y (bp)	21	−40.0	2.2	−45.4	−37.0
Tape swap spread, 30Y (bp)	21	−69.7	2.1	−74.4	−66.5
Naive CDX IG index-basket gap (bp, daily)	21	−33.2	13.8	−56.3	−2.0

Notes: Netting ratio is settled divided by compared volume (Section 6). Swap spreads are median fixed rates on new trades less same-day Treasury par yields (Section 7). The naive index-basket gap compares the CDX NA IG index with a 184-name segment basket and is a composition artefact (Section 13). Source: DTCC files and tape (DTCC, 2026b,a,c); FRED (Federal Reserve Bank of St. Louis, 2026).

5 Framework: enforcement classes, and the state-space form

5.1 Five enforcement classes

Table 3 sorts the catalogue. The class of a relationship, not its asset class, determines its dynamics, and cross-relationship correlation in stress is a property of the class: class-4 deviations share the dealer balance-sheet factor and widen together, while class-2 deviations follow calendars and legislatures and owe nothing to class 4.

Class 1: identities. An identity holds by construction of the records. It cannot be traded and it cannot be violated; an apparent violation is a data error, and locating it is a measurement result. The netting identity (Section 6) and the waterfall leg of the triangle (Section 20) are the catalogue's two identities. Their scientific content is in their interpretation: the netting ratio is a state variable of the cleared system, and the waterfall is an audit that ties three independent data pipelines to each other.

Class 2: contract and statute. A delivery obligation, a funding formula, a capital-control law and a tax exemption each define a boundary and a priced or policed consequence for crossing it. Convergence is calendrical (settlement dates, roll dates, funding intervals, tax years), and the characteristic failure is legislative: the TCJA of 2017 for municipals, the 2023 Nigerian window unification for segmentation, the introduction of the fails charge in 2009 for repo. Backtests across such breaks estimate a law change as if it were a mispricing.

Class 3: physical conversion and replication. A refinery, a power plant and an index replication portfolio convert inputs to outputs at a technical rate. The spread between output and input values is bounded by the conversion technology and by the option to shut down or substitute, and it converges at the speed of physical adjustment (maintenance cycles, capacity additions), not at the speed of arbitrage capital.

Class 4: no-arbitrage with a balance-sheet band. The enforcement trade exists and is well understood, but it consumes balance sheet that is charged by the leverage ratio, by margin and by funding markets. The deviation's equilibrium is the band edge, its width is a price (the shadow rent

Table 2: Data blocks and coverage

Block	Coverage	Size
Swap tape (DTCC PPD)	Sep 2026 full; Jul, Aug, Oct slices	60,245,115 trades
FICC clearing volumes	Jan 2001 to Sep 2026, monthly	1,236 values
GCF index / fails	Oct 2025 to Oct 2026, daily (rolling)	250 days each
Sponsored volumes	Aug 2021 to Oct 2026, daily	1,289 days
Underwriting	Jan 2011 to Aug 2026, monthly	188 months, 4 products
ABS-EE loan tape (registered)	Nov 2016 to Oct 2026 filings	28,638 loans, 517 deals
Form 10-D tranche record	2012 to Sep 2026 statements	738,118 tranche-months
Rating events (17g-7)	To Oct/Nov 2025 (agency files end)	28,292,725 events
N-PORT holdings and marks	28 quarters	612.6M rows, all tables
Bond prints (TRACE store)	707-bond universe	4,100,179 trades

Notes: Retrieval dates and limitations per series are in Appendix C. Source: as cited per block in the text.

on balance sheet), and its dynamics are funding dynamics. Swap spreads, the CIP basis, the funded leg of the cash-futures basis, the CDS-bond basis and index additivity on its funding side all live here, and the joint test in Appendix B is a test of the class, not of any single member.

Class 5: statistical regularities. No contract compels convergence. The convenience yield in its tradable form, the correlation premium, the ratings leg of the triangle and holder-implied pricing are regularities sustained by regulation, clientele and accounting practice. They have premia rather than half-lives, and the correct output of research on them is a decomposition with standard errors, not a trade.

5.2 The state-space form

For every relationship we write the measured deviation as

$$y_t = \mu_t + c_t + e_t, \qquad \mu_t = \mu_{t-1} + \eta_t, \tag{1}$$

where μ_t is the current equilibrium level of the deviation, c_t is the position within the enforcement band, and e_t is measurement error. The textbook relative-value model is the special case $\mu_t = \mu$ constant, c_t an Ornstein-Uhlenbeck position, $e_t = 0$: a fixed target, mean reversion, and a half-life. Most published half-lives are estimates of that special case. The catalogue's histories reject it in both directions: for class-2 and class-4 relationships the equilibrium itself moves (with balance-sheet prices, with statute, with supply), so η_t has material variance; and for several relationships the measurement term e_t is the largest of the three, as Section 13's composition gap shows in pure form. The practical consequence is a reporting rule we apply throughout: every deviation is reported with its μ context (what the level should be, given the costing), its band position, and its measurement audit, and no half-life is quoted for a series whose μ has a known break inside the sample.

5.3 Lead-lag discipline

Claims that one leg of a relationship leads the other are identified only under three restrictions, which we enforce in Sections 13, 20 and 21. Event times must be aligned to the information timestamp (a quarterly fund mark is stamped at its report date; a test that lets it lead a daily print is testing the calendar). Where synchronous data exist, lead-lag is measured by information shares in the

Table 3: The catalogue by enforcement class, with data status

Sec.	Relationship	Class	Data status
6	Clearing netting identity	1 Identity	In hand, 2001-2026
7	Swap spread	4 Balance-sheet band	Measured, Sep 2026
8	CIP / cross-currency basis	4 Balance-sheet band	Feasibility in hand
9	Treasury convenience yield	5 Statistical (institutional)	Decomposition in hand
10	Repo specialness, fails, GCF	2 Contract	Aggregate leg in hand
11	Cash-futures basis	2/4 Contract, funded	Futures leg not held
12	CDS-bond basis	4 Balance-sheet band	CDS leg in hand
13	Index vs constituents	3/5 Additivity, model	Additivity leg in hand
14	Equity correlation premium	5 Statistical	Not measurable here
15	Conversion spreads	3 Physical conversion	Product leg not held
16	TBA dollar roll	2 Contract, statistical core	Not measurable here
17	Perpetual futures funding	2 Contract (rule)	Not measurable here
18	Market segmentation	2 Statute	Offshore leg only
19	Municipal ratio	2 Statute (tax)	Not measurable here
20	Loan-tranche-ratings triangle	1/5 Identity + statistical	In hand, joins built
21	Holder-implied pricing	5 Statistical (accounting)	Panel in hand

Notes: Class 1 identity; class 2 contract and statute; class 3 physical conversion and synthetic replication; class 4 no-arbitrage with a balance-sheet band; class 5 behavioural and statistical regularities. Data status is as of October 2026 and is stated in full in each section.

sense of Hasbrouck, not by cross-correlations of sampled series. And the asynchronous cases are named as such: the cleanest identified lead-lag in our data is the rating leg of the triangle, where the lag is institutional (agencies respond to collateral states with a delay) and the estimand is the lag distribution itself.

5.4 Conservation and network quantities

Two catalogue objects are flows through a network rather than prices. The netting multiplier of Section 6 is the cleared system's analogue of collateral velocity in Singh (2012): Singh's estimates put repledged collateral velocity at 3.0 in 2007 ($10 trillion repledged on a $3.4 trillion source base), 2.4 in 2010 and 1.8 by 2015. The GSD multiplier moved from 3.37 (January 2001) to 3.06 (September 2026) (Observed, A). The two series measure different systems (bilateral rehypothecation versus central netting) but the same quantity: how much gross exposure each unit of settled or sourced exposure supports, and therefore how much gross exposure reappears if the compression mechanism fails.

Optimal transport enters the framework as a metric, not a signal: the cost of transforming one curve or distribution into another is a well-defined distance between market states (yield curves across dates, spread distributions across months), useful for stating how far a regime has moved. It generates no trades, and we use it only descriptively.

5.5 The spuriousness protocol

Every featured relationship passed four screens, in order. (1) A named enforcer exists. (2) A loss state is stated, and any premium is interpretable as payment for bearing it. (3) Multiplicity is accounted: sixteen featured relationships were selected from a larger examined set (Appendix A), so unadjusted

Table 4: Netting by era, GSD monthly averages

Era	Months	Compared ($tn)	Settled ($tn)	ν (%)
2001-2007	84	48.0	15.0	31.4
2008-2009	24	66.5	18.8	28.2
2010-2019	120	68.6	25.7	37.6
2020-2021	24	91.1	34.4	37.9
2022-2026	57	165.4	59.4	36.3

Notes: ν range within eras: 2001-2007, 28.7 to 36.8 percent; 2008-2009, 25.5 to 32.6; 2010-2019, 24.6 to 45.3; 2020-2021, 32.8 to 41.2; 2022-2026, 32.3 to 38.9. Full-sample ν: mean 35.0 percent, standard deviation 4.7 percentage points. Source: DTCC (DTCC, 2026b); computations by the authors.

5 percent testing across the catalogue would manufacture false positives by construction; the three formal tests in Appendix B carry Bonferroni-adjusted thresholds. (4) Parameters estimated across a known regime break are reported as two estimates, never one. Sections 14 and 21 are included with their screen-3 status recorded (documented premium, untestable here; and unresolved by design, respectively), because a catalogue that only prints its successes teaches the wrong lesson.

6 The clearing netting identity

6.1 The identity

For each clearing division and month, define G as compared volume (the gross obligations created by matched trades), N as settled volume (the obligations actually discharged), the netting ratio $\nu = N/G$, and the multiplier $m = 1/\nu = G/N$. The identity is definitional: every compared obligation is either netted away or settled, so $G \equiv N + (\text{netted})$, and ν is the share of gross flow that survives compression. Three economic quantities hang off the ratio. Margin and default-fund requirements scale with N, not G, so ν converts market activity into clearing-house resource demand. The funding footprint of the cleared system (the cash and securities that must move on settlement day) is proportional to N. And the unwind exposure is G: if a participant defaults and its netting set is torn up, obligations re-gross toward G, so m measures the leverage of the netting arrangement itself.

A benchmark disciplines interpretation. If k participants trade in random directions, net positions scale as $\sqrt{k}$ while gross scales as k, giving $\nu \sim 1/\sqrt{k}$: netting efficiency should improve as participation broadens, and deteriorate when flow becomes one-sided or concentrated. Deviations of measured ν from that scaling are information about flow structure, which is why we treat ν as a state variable rather than a technological constant.

6.2 Measurement: twenty-five years of FICC volumes

Figure 1 plots GSD compared and settled volume, January 2001 to September 2026, and Figure 2 the two netting ratios (GSD settled share; MBSD EPN share of clearing par). Table 4 summarizes by era. The anchor points verify the parse: January 2001 compared $28.25 trillion, settled $8.39 trillion, $\nu = 29.7$ percent, $m = 3.37$; September 2026 compared $248.65 trillion, settled $81.29 trillion, $\nu = 32.7$ percent, $m = 3.06$ (OBSERVED, A). Gross volume grew 8.8-fold over the period; settled volume grew 9.7-fold. The system's compression efficiency, relative to its own starting point, fell about 9 percent while its scale rose nine-fold.

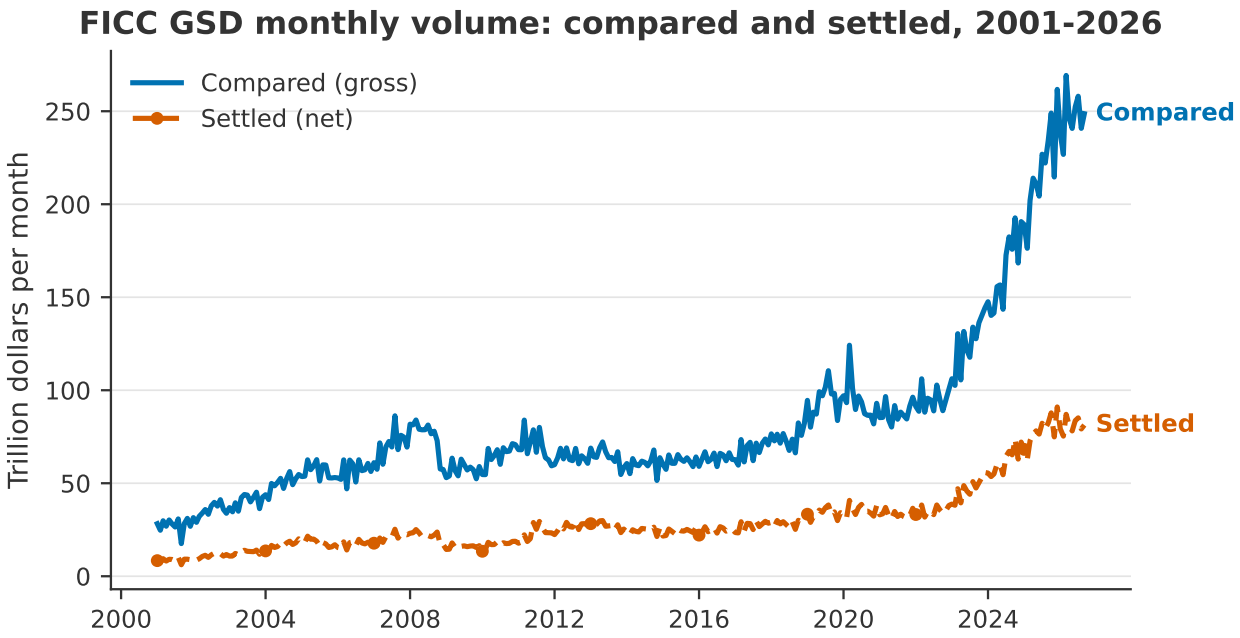


Figure 1: GSD monthly compared and settled volume

Source: DTCC FICC clearing volumes file (DTCC, 2026b). Note: Compared volume is gross obligations created by matched trades; settled volume is the net obligation discharged after netting. Both series are monthly totals.

Three features of the series carry the interpretation. First, ν fell during 2008-2009 (era mean 28.2 percent, the lowest of any era): in the crisis, netting worked harder, because flows were large in both directions across a broad participant set and the clearing house compressed them. The sample minimum, 24.6 percent, falls in January 2010, and the maximum, 45.3 percent, in December 2012, in the post-Dodd-Frank clearing build-out, when newly cleared flow arrived with less offsetting structure. Second, the published settled series shifts level around mid-2011 (settled volume steps from about $18 trillion to about $29 trillion per month between the first and second halves of 2011, with compared volume roughly flat), lifting ν from the high twenties to the high thirties. The file does not document the cause, and we do not assign one; we record it as a series break and treat level comparisons across it as provisional. Third, within the post-2022 era, ν drifts down (2026 average 32.9 percent) even as compared volume roughly doubles from its 2022 level: at the current margin, volume growth is arriving with more offset, not less.

A regression makes the same point conditionally. Regressing ν on log compared volume and a time trend (309 months) gives a log-volume coefficient of -9.0 percentage points (standard error 0.8) and a trend of $+0.85$ percentage points per year (standard error 0.05), with $R^2 = 0.50$ (Reconstructed, B). The two regressors are strongly collinear (volume trends), so we read the pair jointly: the level of ν is not a volume elasticity; it is a slow state variable with a rising trend component (mandates and membership change) and a negative within-regime volume component (bigger months net more).

MBSD tells the complementary story. The EPN share of clearing par averaged 19.2 percent in 2001 and 2.6 percent in September 2026 (2026 average 3.6 percent) (Observed, A): pool-allocated settlement has nearly exited the TBA complex, replaced by netting. The exception proves the rule: EPN current face roughly tripled in spring 2020, from $0.41 trillion in January 2020 to $1.22 trillion in May 2020, while clearing par was flat, taking the share to 16.5 percent in May 2020. In the stress, the allocation channel briefly carried settlement again. September 2026 levels: clearing par $10.19

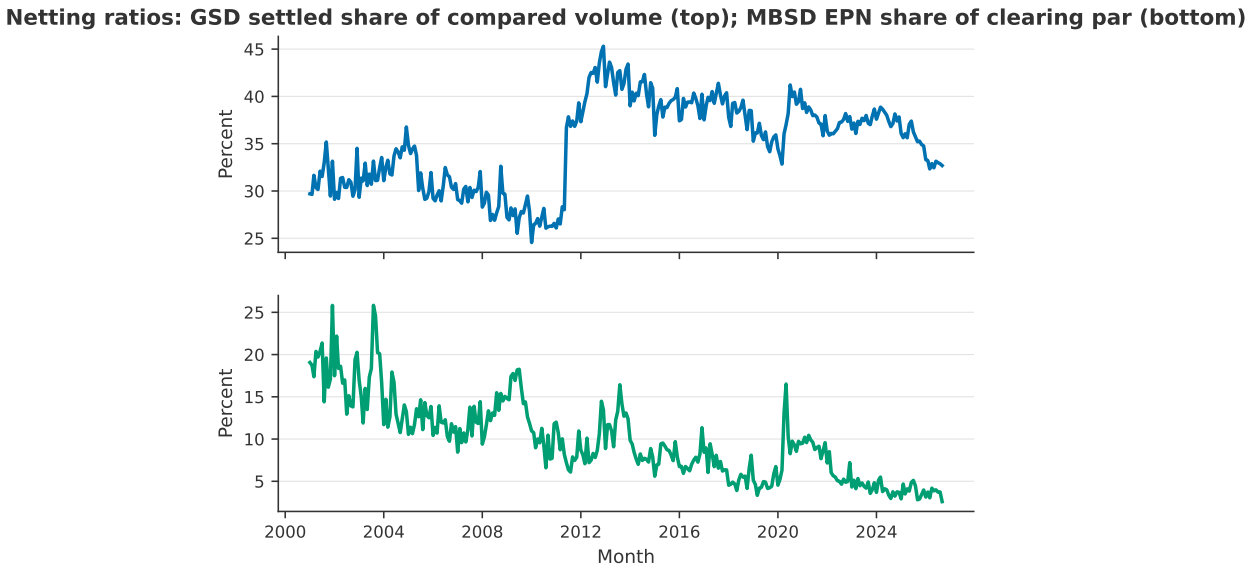


Figure 2: Netting ratios at GSD and MBSD

Source: DTCC FICC clearing volumes file (DTCC, 2026b). Note: Top panel: settled share of compared volume at GSD. Bottom panel: EPN current face as a share of MBSD clearing par. The two ratios measure different settlement technologies and are plotted in separate panels on their own scales.

trillion, EPN current face \$0.264 trillion (OBSERVED, A).

6.3 Convergence, persistence, breakdown

An identity does not converge; it holds. The persistence question is about ν as a state variable: its era means move by 5 to 10 percentage points and stay moved, so any use of ν in margin, funding or capacity models must carry it as a regime variable. The identity's interpretation breaks in exactly one circumstance, participant default, when netting sets are torn up and obligations re-gross toward G; and its measurement is conditional on clearing acceptance, since trades rejected from clearing never enter G. The published ratio is therefore a statement about the cleared population, not about the market's total flow.

6.4 Verdict

Genuine, as an identity and as a state variable; not a trade. Its usable content is second-order and specific: ν should forecast the resource demands of the cleared system (margin, funding, settlement capacity) better than volume does, and Appendix B's Test 2 states the falsifiable version of that claim against settlement fails and GCF pricing. Test 2 has since been assessed: it is not executable as designed, because the fails data its falsifier requires do not exist for the evaluation window, so the claim stands neither supported nor withdrawn (Appendix B).

Table 5: Tape-implied swap spreads, September 2026

Tenor	Days	Trades	Swap rate (%)	Spread mean (bp)	Spread median (bp)
2Y	21	3,625	4.55	−13.3	−13.5
5Y	21	5,831	4.54	−28.1	−27.5
10Y	21	6,316	4.58	−40.0	−39.3
30Y	21	2,839	4.64	−69.7	−69.5

Notes: Dependent measure: daily spread = median fixed rate on new trades less same-day Treasury par yield; table reports the mean and median of the daily spreads and the median swap rate. Within-month standard deviations of the daily spread are 2.1 to 2.8 bp. $N = 21$ business days per tenor. Source: DTCC tape (DTCC, 2026c); FRED (Federal Reserve Bank of St. Louis, 2026).

7 The swap spread

7.1 The relationship and its enforcement trade

The swap spread at tenor T is $s(T) = r_{\text{swap}}(T) - y_{\text{UST}}(T)$, the fixed rate on a par interest rate swap less the Treasury yield of the same maturity. With matched duration, a receive-fixed swap plus a short Treasury is approximately rate-neutral, so in a frictionless market $s(T)$ should sit near the small negative of the dealer's hedging and funding frictions. The enforcement trade for a negative spread $s < 0$ is: buy the Treasury, fund it in repo, pay fixed on the swap. The trade breaks even only if the carry, $-s \times \text{DV01} \times N$ per year, covers the repo funding shortfall on the bond, the initial margin on the cleared swap, and a capital charge ρE on the equity E the position commits under the leverage ratio. The spread therefore has a negative equilibrium whose width is an institution-specific cost stack, and the correct empirical question is not whether s is negative but whether it is negative by more than the stack.

7.2 Measurement on the tape

We compute the spread from the trade tape rather than from a vendor curve. For each business day in September 2026 and each tenor bucket (24, 60, 120, 360 months), we take the median fixed rate on new USD swap trades in the DTCC rates grid, requiring at least ten trades, and subtract the same-day FRED Treasury par yield (DGS2, DGS5, DGS10, DGS30). The construction is a trade-median measure, not a fitted par curve: it prices the swap that actually traded that day, at the cost of mixing package trades and off-run tenors inside each bucket. Table 5 and Figures 3 and 4 report the result (Reconstructed, B).

The spreads are negative at every tenor, monotone in tenor, and quiet: the 10-year daily series runs from −45 to −37 bp with a standard deviation of 2.3 bp. The term structure of the spread (steepening from −13 to −70 bp) is the term structure of the cost stack: longer trades commit more margin, more leverage-ratio exposure and more funding duration per unit of carry.

7.3 The break-even calculation

Calibrate the stack at the 10-year point with October 2026 levels. The Treasury yield is 5.12 percent (20 October 2026 (Observed, A)); the tape median swap rate for September is 4.58 percent, and a par swap rate of 4.82 percent against the October yield gives the round-number spread of −30 bp used in desk discussions. On $100 million notional, DV01 is about $85,000 per bp (10-year annuity factor

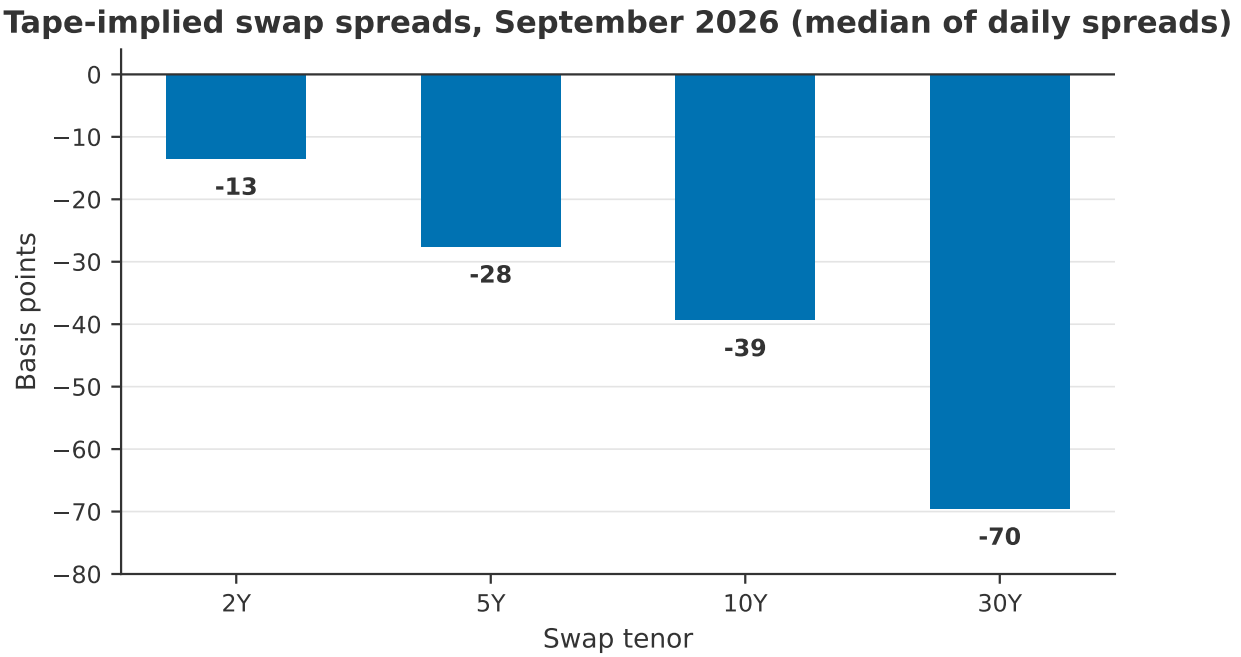


Figure 3: Tape-implied swap spreads by tenor, September 2026

Source: DTCC tape (DTCC, 2026c); FRED (Federal Reserve Bank of St. Louis, 2026). Note: Medians of daily spreads. Bars start at zero; negative values extend downward.

near 8.5), so a −30 bp spread pays about $255,000 per year of carry. Against it: a 2 percent repo haircut commits $2 million of cash, and cleared 10-year initial margin of 2 to 3 percent of notional commits a further $2 to 3 million; at a 10 percent return hurdle on roughly $5 million of committed capital, the annual charge is about $500,000. The trade loses money at −30 bp. It breaks even only when the spread is roughly twice as negative, or when the holder's hurdle and margin treatment are cheaper than a dealer's, which is precisely the configuration the literature describes: the marginal holder is a leveraged non-dealer (a basis fund) whose financing is arranged off the dealer's balance sheet, and the equilibrium spread is set by that holder's stack, not the dealer's (Aquilina et al., 2024; Boyarchenko et al., 2018). Our measured September medians (−39 bp at 10 years, −70 bp at 30 years) sit in the region this arithmetic predicts: beyond a single dealer's break-even at the belly, far inside anything a mispricing reading would need.

The demand side reinforces the costing. Hanson et al. (2022) show that the imbalance between duration supply and receive-fixed hedging demand is itself priced; a month of heavy Treasury issuance with heavy corporate issuance hedging pushes the spread wider (more negative) without any change in the cost stack. The two accounts are complements, and both are stated in levels: neither predicts reversion to zero on any horizon.

7.4 Convergence, persistence, breakdown

Within September 2026 the spread does not converge; it holds a level with a 2.3 bp standard deviation. The relationship's half-life is the half-life of the cost stack and the issuance calendar, measured in quarters. It breaks, in the sense of leaving its band, under three conditions: a funding disruption that raises repo specialness or haircuts (March 2020 is the template), a regulatory change in the leverage ratio or margin schedule (a legislated break, class 2 behaviour inside a class 4 relationship), and a

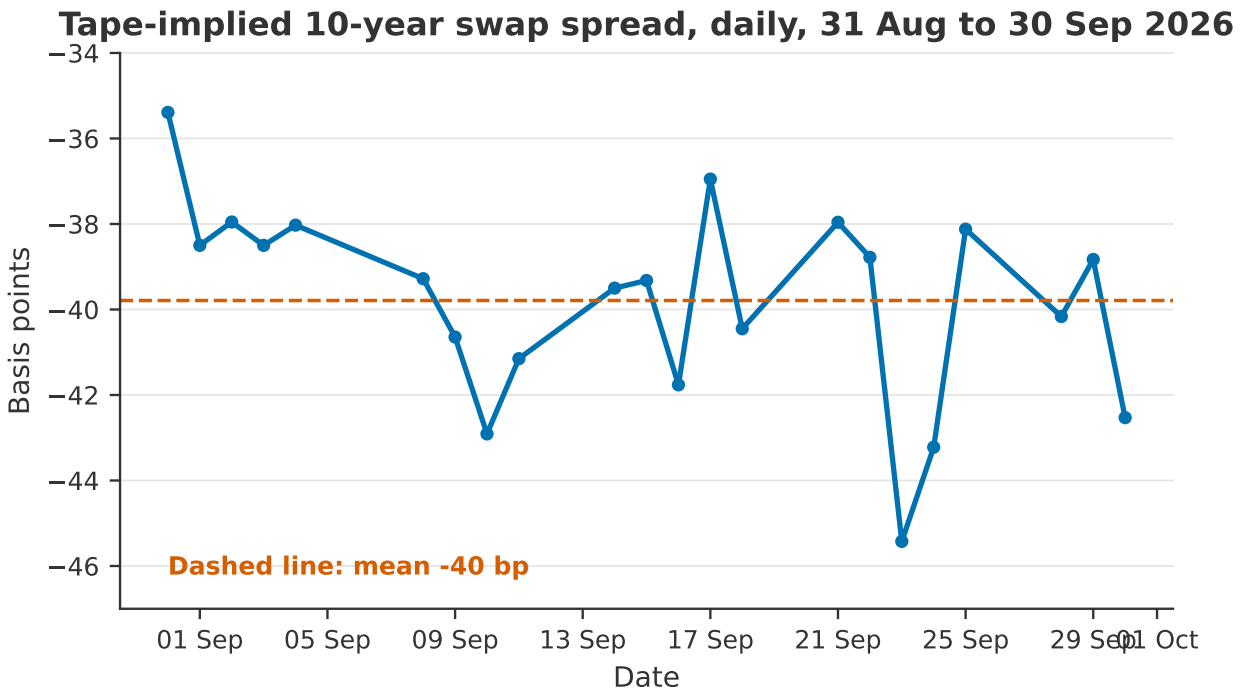


Figure 4: The 10-year tape-implied swap spread, daily

Source: DTCC tape (DTCC, 2026c); FRED (Federal Reserve Bank of St. Louis, 2026). Note: Daily spread, median fixed rate on new 120-month trades less DGS10. The monthly standard deviation is 2.3 bp: in a quiet month the spread is a level, not a trading range.

clearing or settlement failure that reprices the swap leg's counterparty content. The September 2026 configuration (heavy issuance, yields rising on supply, Section 9) is the band's normal operating state, not its edge.

7.5 Verdict

Genuine, as a banded equilibrium. The no-arbitrage statement survives only in its degraded form: the spread equals the marginal holder's balance-sheet cost, plus a demand-imbalance premium, and both terms are measurable in principle. The spurious version of this relationship is the trade that treats any negative spread as an anomaly to be harvested; at September 2026 levels that trade pays about half its capital charge in carry.

8 Covered interest parity and the cross-currency basis

8.1 The relationship

Covered interest parity ties the forward exchange rate to the spot rate and the two money-market rates: $F = S \cdot (1 + (r_{\$} + b)T)/(1 + r_f T)$, where b, the cross-currency basis on the dollar leg, is zero under the textbook parity. A negative b means synthetic dollar funding (borrow foreign currency, swap into dollars) is expensive relative to direct dollar funding. The enforcement trade (borrow foreign, buy dollars spot, sell dollars forward, invest at $r_{\$}$) is balance-sheet intensive in exactly the way

Section 7's trade is: it is a funded, margined position held by an intermediary. The two relationships are therefore one relationship observed in two markets, and we treat them jointly: the swap spread prices the rent on balance sheet committed against Treasury duration; the basis prices the rent on balance sheet committed against dollar funding.

8.2 What the literature measures

Du et al. (2018) remain the reference measurement: one-week Libor bases over 2009-2016 range from −10.4 bp (sterling) to −41.1 bp (Danish krone), with a standard deviation near 25 bp; the euro basis of −18.3 bp makes a $1 billion one-week synthetic dollar borrowing cost about $35,000 per week, or $1.83 million per year if rolled. Their identification uses bonds of the same issuer (KfW) trading in several currencies, so issuer credit is held fixed and the basis is measured clean. Avdjiev et al. (2019) show the deviations across currencies load on a single dollar factor, which is the cross-market form of our unification claim. Cenedese et al. (2019) document the calendar: deviations spike at quarter-ends and year-ends, when G-SIB reporting makes balance sheet most expensive. Central bank swap lines place a policy cap on the basis for currencies that have them, a class-2 boundary inside a class-4 relationship.

The scale of raw rate differentials in October 2026 frames the size of the object: the US 10-year yield is 5.12 percent (20 October (Observed, A)) and the euro-area 10-year AAA spot rate was 3.19 percent at end-June 2026 (ECB, via the Econ Data Library, data_through 2026-06-23). The 193 bp gap between those two numbers is mostly expected policy divergence and term premia, not basis; the basis is the small residual after the forward market has priced the differential, which is why it must be computed from forwards and money-market legs, never from yield gaps.

8.3 Feasibility on the tape

The tape's FX partition for September 2026 holds 1,223,974 new FX trades, and the cross-currency swap population is material at short tenors (14,289 trades, with leg-1 notional concentrated under twelve months) (Observed, A). The basis is in principle computable at daily resolution: the forward leg is in the tape, and the spot and money-market legs come from FRED and the library. Two limitations bind. The tape does not timestamp intraday, so the computation is a daily close-to-close object and cannot see the intraday squeezes where the basis does its damage. And the money-market leg for non-dollar currencies is a library series with publication lags, so a tape-basis series built today is a research construction, not a monitoring feed. These are the same limits, in mirror image, as Section 7's: there the swap leg was in the tape and the bond leg in FRED; here both derivative legs are in the tape and the cash legs are external.

8.4 The joint claim, and its falsifier

Treating Sections 7 and 8 as one relationship generates a prediction that neither section generates alone, and Appendix B states it as Test 3: build the two panels (tape swap spreads by tenor; tape cross-currency bases by currency, with cleared and bilateral trades separated), and the first principal component of the standardized deviations must carry at least 60 percent of common variance with loadings of the predicted sign; the factor must widen at quarter-ends and year-ends; and the widening must be larger in the bilateral subpanel than the cleared one, because clearing mutualizes part of the balance-sheet cost. The cleared-versus-bilateral contrast is the discriminating observation: a pure demand story (issuance, hedging flows) moves both subpanels together, while a balance-sheet story

predicts the split. If the first component carries under 40 percent, or the calendar coefficients are insignificant or wrong-signed, or the contrast is absent, the unification claim is withdrawn and the two relationships return to being separate literatures. The test has been run: the calendar and contrast clauses of the falsifier triggered, and the unification claim is withdrawn (Appendix B).

8.5 Verdict

Genuine, in the same degraded form as the swap spread: a no-arbitrage relation whose deviation is the price of the balance sheet that enforces it. Its persistence is the persistence of that price, its calendar is the reporting calendar, and its breakdown mode is policy (swap lines, leverage-ratio relief), not convergence.

9 The Treasury convenience yield

9.1 The relationship

Write the Treasury yield as $y_{\text{UST}} = y^* - cy$, where y^* is the yield on a hypothetical security with the same cash flows and no special services, and cy is the convenience yield: the non-pecuniary return to holding the asset that is the system's money (collateral eligibility, liquidity on demand, regulatory treatment). Krishnamurthy and Vissing-Jorgensen (2012) estimate cy at about 0.7 percentage points at recent debt-to-GDP ratios, worth roughly 0.3 percent of GDP per year to the issuer, and decreasing in supply. The hedging-perspective manuscript (NYU Stern School of Business, n.d.) adds the state dependence: convenience is high when Treasury returns covary least with equities, that is, when the safe asset hedges best.

Convenience is the upstream variable of this catalogue. It is one component of the swap spread (a convenient bond trades rich, pushing s down), of the on-the-run premium, of the TIPS liquidity residual, and of agency and supranational spread residuals. That generates a testable restriction: after scaling, those residuals should be close to rank one, a single factor with loadings set by each market's convenience sensitivity. The restriction is what makes convenience a relationship rather than a residual label: if the residuals do not co-move, there is no single convenience factor and the concept is doing no work.

9.2 A measured decomposition: September to October 2026

The autumn 2026 window gives the factor a natural experiment. Between 1 September and 20 October 2026 the 10-year nominal yield rose 33 bp (4.79 to 5.12 percent) while the 10-year breakeven inflation rate was unchanged to within 2 bp (2.35 percent on 1 September; 2.35 percent on 20 October; 2.37 percent on 21 October) (Observed, A). The implied real yield therefore rose by the full 33 bp (2.44 to 2.77 percent). Figure 5 plots the two series. A nominal rise with a flat breakeven is the signature of a real, supply-side repricing (term premium and absorption), not of inflation expectations: the joint prediction of the convenience account is that convenience-sensitive residuals (swap spreads wider in the negative direction, on-the-run premia compressed, agency residuals wider) move together in the same window. Our tape swap spreads are measured for September only, so the joint test awaits the October slice; the decomposition itself is the result we report here, and it disciplines the reading of Section 7: the September spread levels were measured in a market already repricing real yields upward on supply, which is the state in which the absorption account expects spreads at their widest.

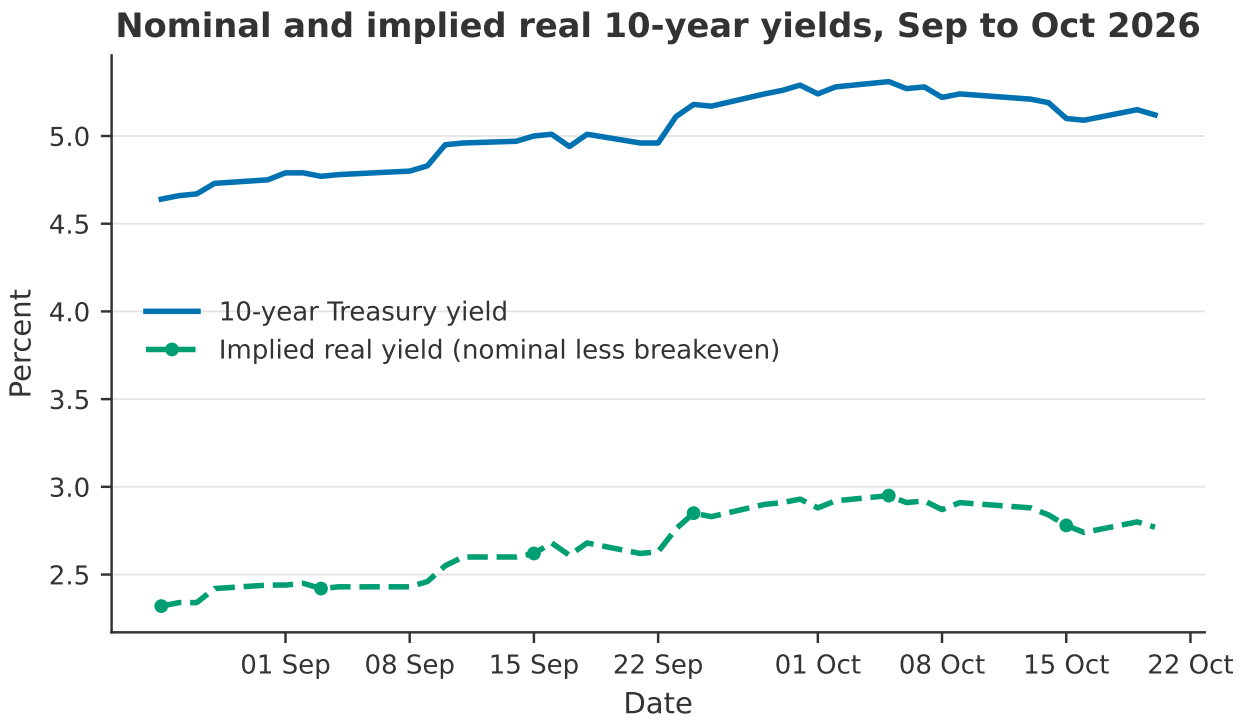


Figure 5: Nominal and implied real 10-year yields, September to October 2026

Source: FRED series DGS10 and T10YIE (Federal Reserve Bank of St. Louis, 2026). Note: Implied real yield is the nominal yield less the breakeven inflation rate. The nominal rise over the window is entirely a real-yield rise.

9.3 Tradability, persistence, breakdown

Convenience is not directly tradable: there is no portfolio that isolates *cy* from term premium, liquidity and safety demand, and every attempted isolation inherits the other terms. Its regime variable is supply: convenience falls as debt-to-GDP rises, on the Krishnamurthy and Vissing-Jorgensen (2012) estimates, so the factor has a slow downward drift under current fiscal paths, punctuated by flight-to-safety spikes. The institutional hardening matters for persistence: liquidity-coverage rules and collateral eligibility schedules write convenience into regulation, which makes part of *cy* a policy parameter rather than a market price, and that part breaks only legislatively.

9.4 Verdict

Genuine as a factor; spurious as a raw trading signal. The one-factor restriction across convenience-sensitive residuals is the correct empirical object, it is computable on our data once a second month of tape slices exists, and until that test is run the factor is a hypothesis with a decomposition, not a result.

10 Repo specialness, settlement fails and the GCF index

10.1 The relationship

A security trades *special* when its specific-collateral repo rate r_i falls below the general collateral rate r_{GC}: $sp_i = r_{GC} - r_i \geq 0$. The lender of cash against that specific security accepts a lower return because the borrower of the security needs it to make a delivery. Duffie (1996) showed the specialness is capitalized into the bond's price, $P_i = P^* + E\left[\int sp_i(t) \cdot d(t)\, dt\right]$ over the expected special period: an on-the-run bond is expensive by the present value of its expected specialness. The enforcement is a delivery contract with a priced escape: since May 2009 the fails charge (3 percent per annum) caps specialness, because failing to deliver costs at most the charge, so sp_i cannot persist much above 3 percent annualized. The auction cycle drives the supply side: Cahill et al. (2013) find supply effects on specialness strongest in auction-cycle weeks three and four, and Cahill et al. (2018) estimate the special collateral premium at about 83 percent of the 10-year on-the-run premium and about 60 percent of the TIPS liquidity residual. Central bank holdings work the other side: Ampudia et al. (2025) estimate Eurosystem purchases depress Bund repo rates about four times more per unit of free float than debt-manager purchases.

10.2 What we can measure: the aggregate leg

Issue-level specific-collateral rates are not in our data; what we hold is the aggregate complex (GCF index rates and pars, daily fails, sponsored volumes), and the aggregate leg answers the aggregate question: is collateral scarce in general, or only in specific issues? Figure 6 plots the two daily series. Over the 250 trading days to October 2026, the GCF weighted average rates are 3.75 percent (Treasury collateral) and 3.77 percent (MBS collateral), with par amounts averaging $62 billion and $92 billion respectively (Observed, A). The MBS less Treasury rate gap averages 1.8 bp with a standard deviation of 1.8 bp, a minimum of −3.5 bp (9 April 2026) and a maximum of 10.2 bp (26 November 2025); on the last observation (16 October 2026) MBS par is $93.58 billion at 3.913 percent and Treasury par $68.73 billion at 3.917 percent, a gap of −0.4 bp (Observed, A). A two-basis-point average gap with a two-basis-point standard deviation is the signature of no aggregate scarcity: general collateral in the two big classes prices almost identically, and the GCF complex is not rationing.

Fails tell the same story with a tail. Treasury fails average $30.9 billion per day (standard deviation $13.0 billion; median $28.6 billion) over the window, against GSD compared volume of about $250 trillion per month: on the last observation (20 October 2026) fails are $30.86 billion, about 0.012 percent of the month's compared volume (Observed, A). Agency fails are vestigial ($8.0 million on 20 October 2026; window mean $0.03 billion). But the distribution has episodes: fails spike to $116.0 billion on 30 March 2026, with secondary peaks near $105 billion in December 2025 and $101 billion in September 2026, the latter two in the autumn escalation window and the March peak in the spring 2026 funding stress. The episodes decay in days. That is the shape the fails charge is designed to produce: fails as a brief, priced pressure valve, not a cascade.

Sponsored volumes (Figure 7) are the structural backdrop: the cleared, sponsored channel through which non-dealer repo now runs grew from $0.16 trillion per day (2021 average) to $2.44 trillion (2026) (Observed, A), a fifteen-fold rise that is the plumbing counterpart of the balance-sheet story in Sections 7 and 8: repo intermediation is migrating to the structure that economizes on dealer balance sheet.

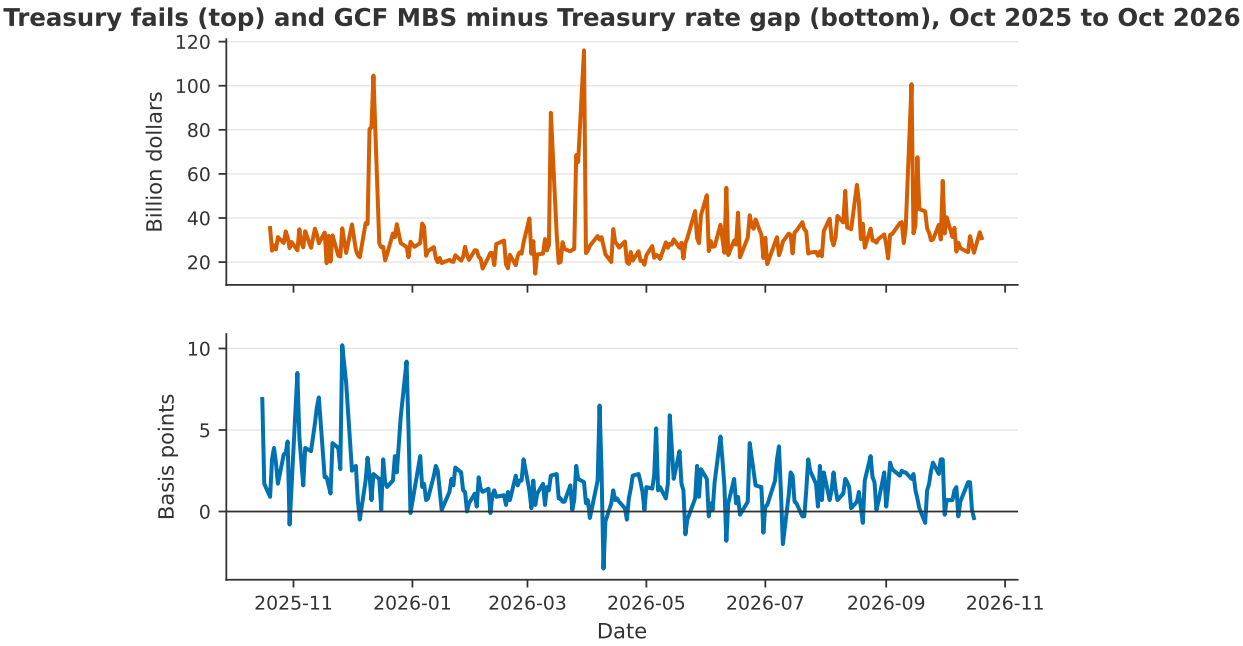


Figure 6: Treasury settlement fails and the GCF collateral gap

Source: DTCC daily fails and GCF files (DTCC, 2026a). Note: Top: Treasury fails to deliver, $ billion per day. Bottom: GCF weighted average rate on MBS collateral less Treasury collateral, basis points. Both files are rolling 250-day windows maintained by the source.

10.3 Convergence, persistence, breakdown

Specialness at the issue level decays with the issue's seasoning and supply events (auction reopenings, buybacks), on a horizon of weeks; the aggregate quantities here do not converge to anything, they diagnose. The relationship breaks when fails stop being priced and start being strategic or operational (a settlement-system outage, a sanctions freeze), and the fails charge itself is a legislated parameter: its level, set in 2009, is the cap in the Duffie formula, and a change in it would reprice every special bond mechanically.

10.4 Verdict

Genuine at the issue level (contractual, with a priced escape), and measurable here only at the aggregate level, where the answer for October 2026 is unambiguous: no aggregate scarcity, gap -0.4 bp on the last observation, fails at 0.012 percent of cleared flow. The spurious version is the inference from aggregate fails to issue-level specialness; the aggregate can be calm while a single issue is cornered, and our data cannot see the corner.

11 The Treasury cash-futures basis

11.1 The relationship

For a deliverable bond and its futures contract, define the net basis as $\text{Basis}_{net} = (P_{cash} + AI) - F \cdot CF - \text{carry}$, where CF is the conversion factor and carry is coupon income less financing cost over

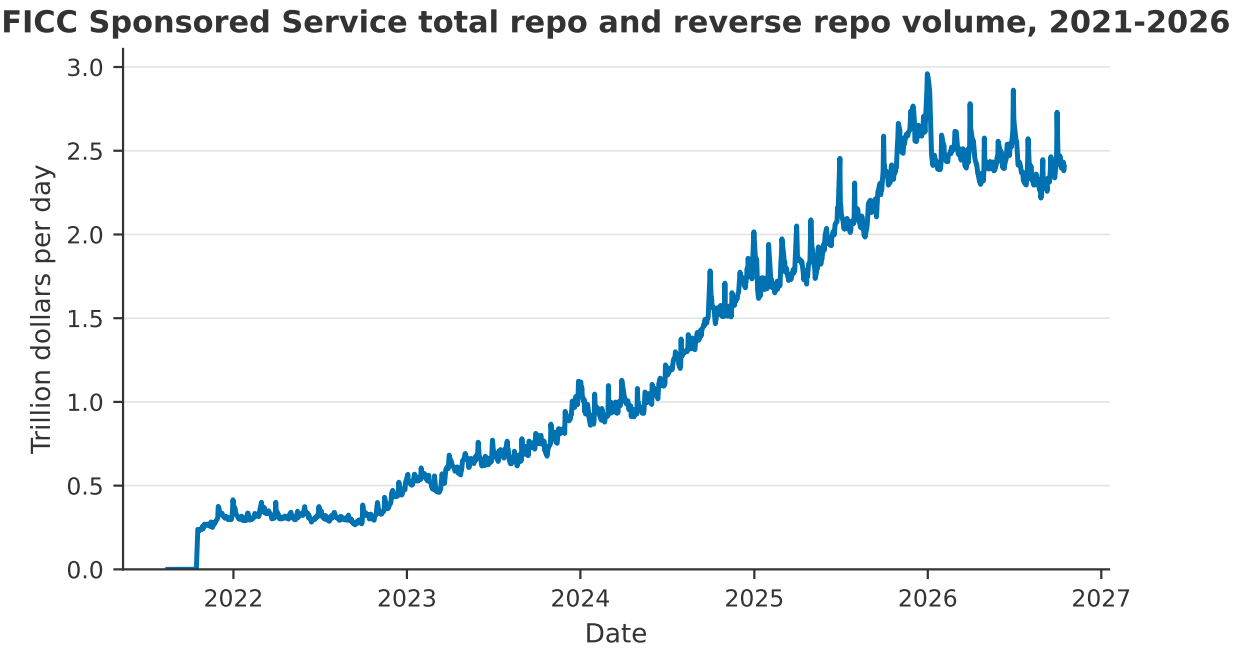


Figure 7: FICC Sponsored Service daily volumes

Source: DTCC Sponsored Service file (DTCC, 2026a). Note: Total repo plus reverse repo, trillion dollars per day. The file's earliest rows (August to October 2021) report zero volume; recorded volumes begin in October 2021.

the holding period. The implied repo rate r^* is the financing rate that sets the net basis to zero. Holding the bond financed at the actual repo rate r_{repo} and short the future earns $r^* - r_{repo}$: the trade is a pure financing arbitrage whose convergence is contractual at delivery. It is class 2 on its convergence leg and class 4 on its funding leg, and the funding leg is the one that matters, because the position is typically run at high leverage by non-dealers and financed in the same repo market whose scarcity Sections 7, 8 and 10 price.

11.2 Scale and the breakdown template

The Federal Reserve's Financial Stability Report of May 2026, as reported in market coverage, puts hedge fund long Treasury exposure intermediated through the basis at about $830 billion as of September 2025, roughly twice the early-2020 peak (Board of Governors of the Federal Reserve System, 2026). Barth and Kahn (2021) document the unwind pattern of March 2020: repo rates and haircuts moved, margin called, and the basis widened violently instead of converging, because convergence required funding that the stress had withdrawn. That episode is the breakdown template for every funded relationship in this catalogue, and it is why the basis belongs in a paper about enforcement: its enforcement mechanism is a contract, but its enforcement *capital* is a funding line.

11.3 A worked example at current rates

Take a bond at 100 with a 4 percent coupon, financed at 4.30 percent for 60 days. Financing cost is 0.717 points; coupon accrual over the period is about 0.667; net carry is −0.05. With a conversion-adjusted futures price of 99.80, the gross basis is 0.20 and the net basis 0.15 points, equivalent to about 90 bp annualized of implied repo advantage over the financing rate (calibrated example; (MODELLED,

B)). Now stress the funding: a 150 bp rise in the repo rate for the final 30 days costs 0.125 points, five-sixths of the net basis. At 50 times leverage, that single funding move is a 6 percent loss on equity, against an expected convergence profit of 7.5 percent if the trade completes. The example uses round numbers, but the asymmetry is structural and does not depend on them: the convergence profit is capped by the basis, the funding loss is not capped, and the ratio of the two is set by leverage.

Futures prices and conversion factors are not in our data vault, so the basis itself is not computed in this paper; the example is calibrated, and the relationship is documented-but-untested here. We flag one measurable implication for the data we do hold: basis-unwind episodes should appear in our series as simultaneous spikes in fails (Section 10), sponsored volume dislocations, and gross GSD volume with a deteriorating netting ratio (Section 6), and the March 2026 fails spike ($116.0 billion on 30 March 2026 (OBSERVED, A)) is a candidate observation for that joint signature, recorded here as a conjecture to be tested, not a finding.

11.4 Verdict

Genuine: contractual convergence with a funding fuse. Spurious only in its marketing form, in which the implied repo advantage is quoted without the leverage and funding terms that convert a 15-point basis into a coin-flip on equity.

12 The CDS-bond basis and the single-name credit panel

12.1 The relationship

Under the credit triangle, a CDS spread s prices default intensity λ and recovery R as $s \approx \lambda(1 - R)$. A par bond of the same issuer prices the same default risk in its spread z over the risk-free curve. The basis $b = s - z$ should therefore be near zero, adjusted for the cheapest-to-deliver option embedded in the CDS. In practice $b \approx -(\text{funding wedge} + \text{liquidity premium}) + \text{CTD value}$: the bond leg must be funded and held, the CDS leg must be margined, and the difference between those two balance-sheet treatments is the basis. Bai and Collin-Dufresne (2011) measure the investment-grade basis near +5 bp before the crisis and near −250 bp inside it, with high yield near −650 bp; Augustin and Schnitzler (2020) show the funding leg carries the explanatory power, with funding illiquidity reinforcing asset illiquidity in the cross-section. Our convention: negative basis means the bond is cheap relative to protection.

12.2 The CDS leg, measured

The September 2026 tape gives the CDS leg at scale (Section 4). Table 6 and Figure 8 report the sector baskets: median member five-year spreads by sector, with member counts. Two measurement facts govern everything downstream. Membership is thin and churns (84 members in July, 103 in August, 127 in September, 75 in the partial October slice), so month-to-month basket moves mix price changes with composition changes; September, the full month with the most members, is the reference. And the basket distribution is extremely skewed: across the 127 September members the median spread is 59 bp, the 90th percentile 130 bp, and the maximum 598 bp (Table 1; Figure 9).

Table 7 converts the widest spreads through the credit triangle at 40 percent recovery. Argentina at 598 bp implies a risk-neutral default intensity of about 10.0 percent per year and a five-year risk-neutral default probability of 39.2 percent; Oracle at 205.5 bp implies 3.4 percent per year and

Table 6: CDS sector baskets: median member 5-year spread (bp)

Sector	Jul 2026	Aug 2026	Sep 2026	Oct 2026	Members, Sep
Industrials	38.1	25.0	29.8	95.0	14
Healthcare	45.5	30.0	40.0	31.5	13
Utilities and Materials	36.5	39.8	43.8	49.1	15
Energy	135.8	49.2	53.7	116.0	11
Consumer	81.5	52.0	59.0	78.9	19
Financials	56.5	58.2	63.6	66.2	16
Telecom and Media	61.0	63.0	65.0	73.5	9
Technology	68.0	73.0	77.9	73.5	16
Sovereigns	125.0	81.8	105.5	95.0	10

Notes: Basket spread is the median of member median 5-year spreads; membership requires at least three priced 5-year trades in the month. October 2026 is a partial slice through 19 October with 75 members and is not comparable in membership to September. Source: DTCC tape (DTCC, 2026c); sector assignments are the authors' hand classification of 272 entities.

Table 7: Widest single-name 5-year CDS spreads, September 2026

Name	Sector	Spread (bp)	Trades	λ (%/yr)	5Y PD (%)
Argentine Republic	Sovereigns	598.0	50	10.0	39.2
Republic of Ecuador	Sovereigns	400.0	5	6.7	28.3
Paramount Global	Telecom and Media	327.0	4	5.4	23.9
CCO Holdings	Telecom and Media	306.5	4	5.1	22.5
DXC Technology	Technology	238.0	12	4.0	18.0
Petroleos Mexicanos	Energy	233.0	45	3.9	17.6
Oracle Corporation	Technology	205.5	84	3.4	15.7
Stellantis	Consumer	173.0	3	2.9	13.4
Republic of Colombia	Sovereigns	147.5	125	2.5	11.6
Ford Motor Company	Consumer	143.5	20	2.4	11.3

Notes: $\lambda = s/(1-R)$ at $R = 40$ percent; 5Y PD $= 1 - e^{-5\lambda}$, risk-neutral. Trades are priced 5-year trades in the month. Source: DTCC tape (DTCC, 2026c).

15.7 percent over five years (Reconstructed, B). These are risk-neutral objects (they contain the default risk premium), and we report them as the tape's own arithmetic, not as forecasts.

12.3 The basis itself: what is computable and what is not

The bond leg joins to the CDS leg only for names that are both in the priced CDS population and in the TRACE print universe, and that join has not been built: it is a separate construction (the mixed-data build), deliberately kept out of the swaps-only measurement layer. What we can do here is fix the decomposition on a worked example and state the population boundary. Example: a name with CDS at 205.5 bp and a cash spread of 260 bp has a basis of −54.5 bp; the CTD option is worth perhaps +5 to +20 bp; the funding and liquidity residual is then −65 to −75 bp, which at September 2026 financing costs is the order of magnitude the funding account predicts for a five-year funded position (calibrated example; (Modelled, C)). The boundary matters as much: the widest credit story in our monitored universe, CoreWeave, is not in the CDS priced population at all, so no basis

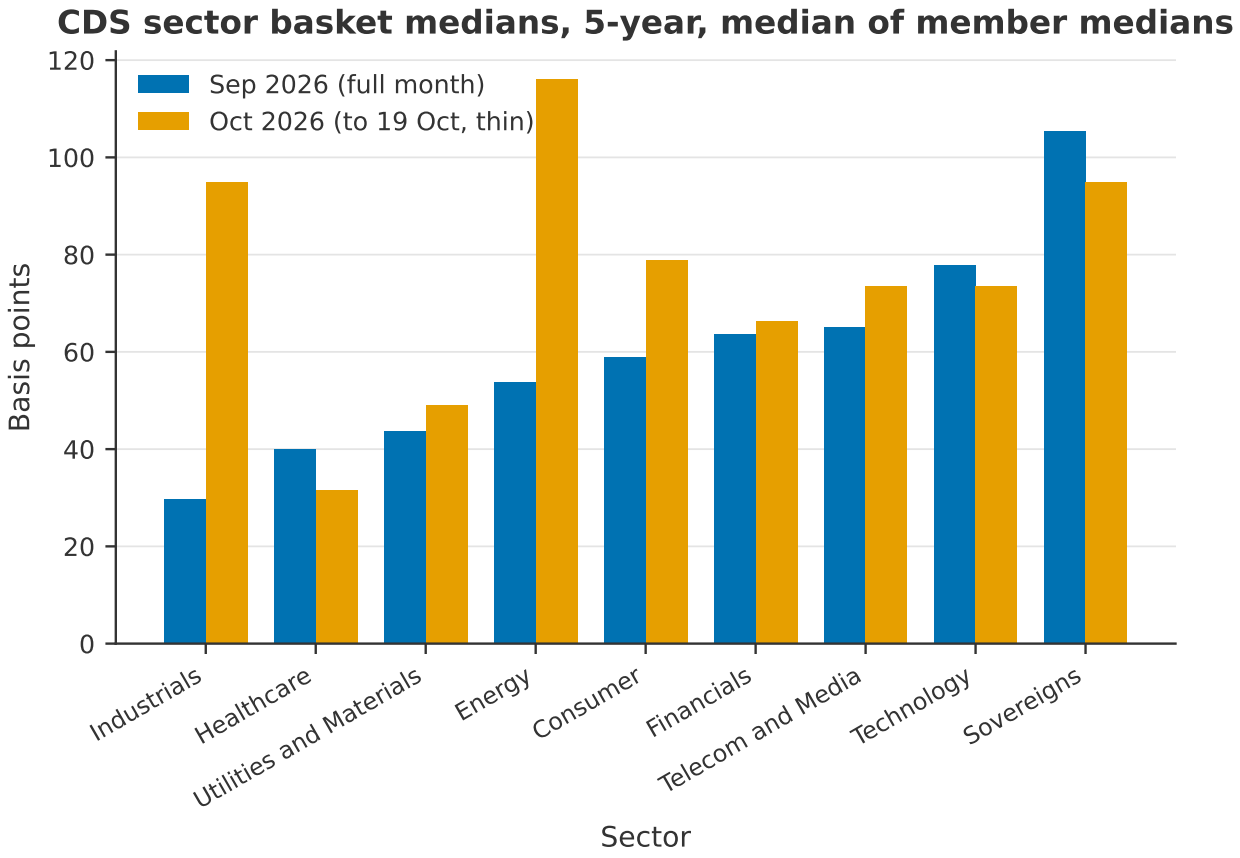


Figure 8: CDS sector basket medians, September and October 2026

Source: DTCC tape (DTCC, 2026c). Note: October is a partial, thinner slice (75 members against 127); the Industrials and Energy gaps between the months are substantially composition (single wide names entering the October membership), which is the point the text makes about basket construction.

statement about it can be made from the tape. Its senior bonds' monthly last prints fell through 2026 (for example CUSIP 21873SAB4: 102.24 in May, 100.72 in June, 93.60 in July, 92.98 in August, 93.45 in September, 85.05 on 20 October (OBSERVED, A)), a cash-market fact with no CDS counterpart in the public record. Press-reported CDS levels for the name (near 1,194.5 bp intraday on 15 October 2026, as reported) are not tape observations and are not used.

12.4 Convergence, persistence, breakdown

The basis is banded on both sides: the negative side by funding and balance sheet, the positive side by the difficulty of shorting cash bonds and by CDS delivery mechanics. Its persistence is funding persistence. It breaks at default events (the basis jumps to the recovery arithmetic), at CDS contract events (succession, restructuring determinations), and in the funding stress of the March 2020 type, when the band widens by an order of magnitude, as Bai and Collin-Dufresne (2011) document for 2008-2009.

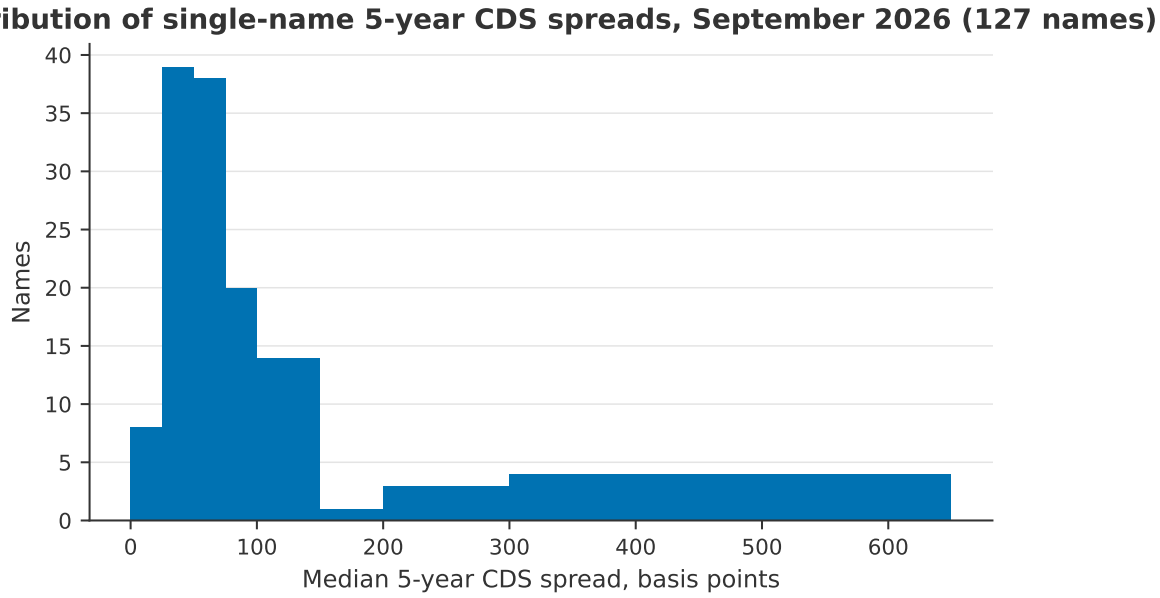


Figure 9: Distribution of single-name 5-year CDS spreads, September 2026

Source: DTCC tape (DTCC, 2026c). Note: Median spread per name over the month, 127 names with at least three priced 5-year trades. Bins are unequal at the right tail by construction (200-300 and 300-650 bp).

12.5 Verdict

Genuine, class 4, with the funding leg dominant. In our data the CDS leg is measured and the basis awaits its join; the honest empirical content of this section is the panel, the triangle arithmetic, and the population boundary, and any basis number quoted without the join's coverage statement should be read as marketing.

13 Index versus constituents, and base correlation

13.1 Two restrictions, not one

A CDS index bundles n names (125 for CDX North America Investment Grade) at equal notional weights. Two distinct restrictions follow. The *additivity* restriction: the index spread should equal the risky-duration-weighted average of constituent spreads, up to funding and liquidity terms, because the index is replicated by the constituent portfolio. This restriction is enforced by replication and belongs to class 4 (with class 3's replication logic). The *tranche* restriction: index tranche prices imply, through a copula, a base correlation curve $\rho_{base}(d)$ defined recursively by $V(0, d; \rho_{base}(d)) =$ the market's cumulative tranche value to detachment d. The implied skew ($d\rho_{base}/dd > 0$) has been documented since Bank for International Settlements (2005). This second restriction is model-relative, not arbitrage-enforced: base correlation is a quoting convention for a model, and its deviations are statements about the model, not about prices. Keeping the two restrictions separate is the whole content of this section's verdict.

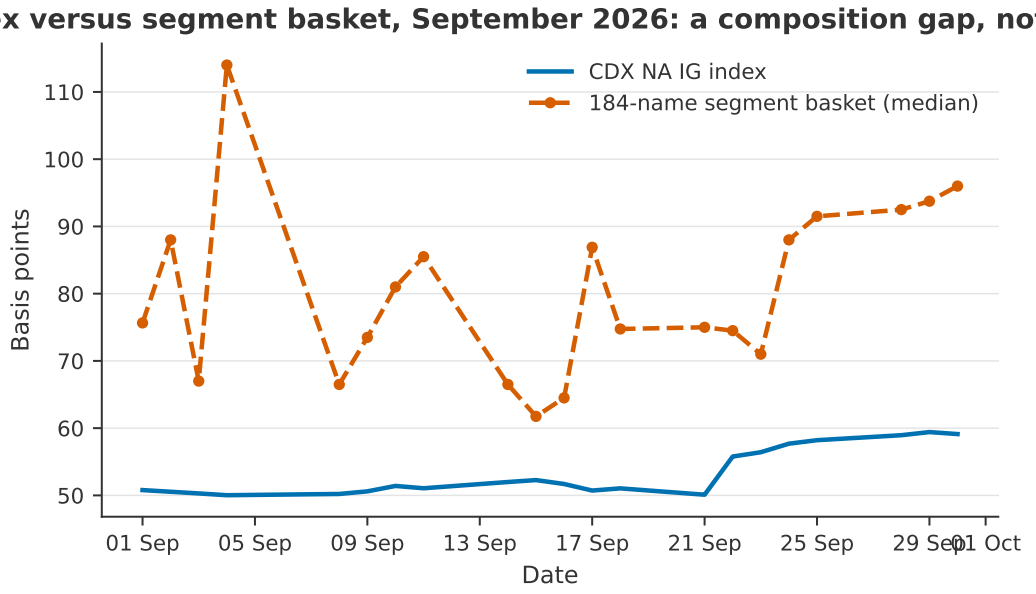


Figure 10: CDX NA IG index versus the segment basket, September 2026

Source: DTCC tape (DTCC, 2026c). Note: The basket is the 184-name investment-grade segment population, not the 125 index constituents. The persistent gap is composition, and the figure is included as the paper's worked example of a spurious basis. Lines connect consecutive trading days.

13.2 The additivity leg, and how to fake it

The tape supports the additivity computation: index trades carry the index factor on 100 percent of trades, six launch indices print, and CDX NA IG five-year daily medians for September 2026 run 50.0 to 59.4 bp with a median of medians of 51.4 bp over 21 days (OBSERVED, A) (the step up in the final week is the September roll, when the constituent set refreshes). What the tape does not carry is the index constituent list joined to the single-name panel, and the distance between the correct computation and the naive one is the paper's cleanest spuriousness exhibit. Figure 10 and Table 1 report the naive version: the index against the median of the 184-name IG *segment* basket the statistics layer builds. The gap averages −33.2 bp with a standard deviation of 13.8 bp across the 21 days (RECONSTRUCTED, B). It looks like a rich, stable basis. It is composition: the segment basket contains sovereigns-adjacent and non-constituent names with spreads far above the investment-grade constituent set (the September member distribution runs to 598 bp), and a median over that population sits permanently above the index. The high-yield version is worse: a nine-name HY segment basket against CDX HY gives a naive gap averaging −209.9 bp with a standard deviation of 102.3 bp. Tightness is not evidence: the IG gap's t-statistic against zero is about −11, and it is entirely artefact. Any screening procedure that ranks bases by statistical significance will promote this one to the top of the list, which is why the catalogue's screen 1 (a named enforcer operating on the correct population) precedes all statistics.

The correct computation is specified, and awaits the constituent join: for each day, weight the 125 constituent five-year spreads by risky duration, compare with the index print, and attribute the residual to funding (class 4) and to index liquidity. The September roll is a built-in control: the constituent set changes on a known date, so composition effects and price effects are separable around it.

13.3 The tranche leg

Index tranche quotes are not in the vault and are not public at trade level; the dissemination tape does not carry tranche prices. The base-correlation section of this relationship is therefore documented, not measured: the skew exists, its arbitrage content is nil (there is no trade that enforces a copula), and its research content is model diagnostics. We record the data status plainly: unavailable, with no proxy.

13.4 Verdict

Split verdict. Additivity: genuine, class 4, computable on the correct population, and the naive computation is the catalogue's canonical spurious result. Base correlation: a model quotation device, genuine as diagnostics, spurious as arbitrage, and untestable on public data.

14 Equity dispersion and the correlation risk premium

14.1 The relationship

Index variance decomposes exactly:

$$\sigma_I^2 = \sum_i w_i^2 \sigma_i^2 + \sum_{i \neq j} w_i w_j \sigma_i \sigma_j \rho_{ij}. \tag{2}$$

Selling index variance and buying constituent variance (a dispersion trade) is therefore a sale of implied correlation against realized correlation. Driessen et al. (2009) estimate the wedge for the S&P 100 over 1996-2003: implied correlation averages 46.7 percent against realized 28.7 percent, an 18-point premium. Bossu (2014) give the trade's mechanics: dispersion profit and loss is proportional to $(1 - \rho_{realized})$ times variance notional, so the position collects steadily while correlation is ordinary and loses, without bound at the portfolio level, in the state where correlation goes to one (a market crash, in which every constituent falls together and the short index-variance leg's hedge value inverts).

A stylized calibration fixes the scale: on $100 million of variance notional per leg at a variance scale of 0.04, the Driessen et al. (2009) wedge is worth about $720,000 of expected collection per period (Modelled, C). The number is an upper bound in a specific sense we state plainly: the 1996-2003 estimation window contains exactly one full correlation event, so the premium estimate is identified off a sample in which the loss state occurred once and mildly.

14.2 Data status: not measurable here

Two independent absences block measurement. We hold no listed option prices, so neither implied constituent variance nor implied index variance exists in the data. And the DTCC equity tape does not rescue the question from the other side: the underlier identity field is populated on 0.0 percent of the 7.41 million new equity trades in September 2026 (Observed, A), so even realized constituent correlation cannot be tied to named underliers from the tape. The section is framework-only by necessity, and we classify the relationship's screen-3 status accordingly: documented premium in the literature, untestable on current data, included in the catalogue because any future options-data acquisition should be evaluated against this section's design.

14.3 Convergence, persistence, breakdown

There is no convergence: implied correlation is a price, and the premium persists as long as investors pay for crash insurance in index form. The trade's breakdown is its design: the loss state is the portfolio's worst state, margin is called against the position exactly when correlation spikes, and the position is short the same convexity that the rest of a trading book is long. A premium whose loss state coincides with the firm's stress state is not diversifiable income; it is a leveraged sale of insurance on the firm's own funding.

14.4 Verdict

Genuine premium; spurious as arbitrage. The 18-point wedge is compensation for correlation-event risk, and any presentation of it as a pricing error fails the catalogue's screen 2 (a stated loss state).

15 Physical conversion spreads: crack, spark and input-output systems

15.1 The relationship

A conversion spread prices a technical transformation. The 3:2:1 crack spread, $C = (2P_{RBOB} + P_{ULSD} - 3P_{crude})/3$ per barrel (gallon prices converted at 42 gallons per barrel), is the gross margin of a representative refinery. The spark spread, $S = P_{power} - H \cdot P_{gas} - VOM -$ carbon cost, is the gross margin of a gas-fired plant with heat rate H. In matrix form both are Leontief relations $y = Ax$: fixed input requirements per unit of output. The enforcement is physical: if the spread exceeds the full cost of conversion including the return on the plant, capacity enters or restarts; if it falls below variable cost, the plant shuts and the shutdown option bounds the spread from below. Convergence operates at the speed of physical adjustment (maintenance schedules, permitting, construction), which makes class 3 the slowest of the enforced classes and the one whose bands are widest in the short run.

15.2 The autumn 2026 configuration

The October 2026 market gives the relationship a live calibration. CrossVol Research's October 2026 newsletter, as reported, records the December 2026 diesel crack against Brent peaking near $97 per barrel in September 2026, with the 3:2:1 crack above $50 for months, on lost refining capacity and the closure of the Strait of Hormuz to normal traffic. Against the verified crude price (WTI $107.89 on 20 October 2026 (Observed, A), FRED DCOILWTICO), a diesel crack of $88 to $97 implies diesel at $196 to $205 per barrel equivalent, or $4.67 to $4.88 per gallon, against crude at $2.57 per gallon. The conversion margin is then about 45 percent of the delivered product value. In this configuration the crack is not a statistical spread: it is the scarcity rent of refining capacity, and a position long crude against products is a short position in that capacity, with the loss state being a capacity return (a refinery restart, a strait reopening) that reprices the margin discontinuously.

15.3 Data position

Our tape holds the derivatives shadow of this market, not the market: the DTCC commodities partition records 112,513 CFTC commodity swap trades in September 2026 (Observed, A), which are swaps on commodity underliers, not the futures prices that define the crack. The defining series

(product futures, power prices) come from exchanges and from EIA publications; the EIA series in the Econ Data Library carry a data_through of 2026-05-26, too stale for the autumn configuration, so the calibration above rests on the verified FRED crude price and the reported crack range, and is graded accordingly. The measurement design is standard (daily futures curves for crude, RBOB and ULSD; the spread computed on matched contract months; capacity utilization as the regime variable) and is deferred to a data acquisition we have not made.

15.4 Verdict

Genuine, class 3: the strongest enforcement in the catalogue after identity, and still a band, whose width is set by the cost and the shutdown option of the marginal plant. The spurious version is the mean-reversion trade on the spread: a capacity rent has no mean to revert to while the capacity is out.

16 The TBA dollar roll

16.1 The relationship

A dollar roll sells a TBA position for near settlement and buys it back for a later month. The price drop between the two settlements, against the coupon and paydown income forgone over the roll period, defines an implied financing rate: $P_1(1 + r_{IFR} \cdot d/360) = P_2 + C + \text{paydowns}$. Roll *specialness* is $r_{repo} - r_{IFR} > 0$: when the front month is scarce (heavy CMO issuance demand for collateral, a squeezed coupon), the implied financing rate falls below, and can fall far below, the repo rate, because holding the front contract is worth more than its cash flows. Kitsul and Ochoa (2016) build a funding-liquidity factor from roll implied financing rates and show it is priced in the agency MBS cross-section; Song and Zhu (2019) characterize specialness directly, linking it to the value of the cheapest-to-deliver option, dealer leverage and prepayment exposure. The relationship is class 2 (a contractual exchange over the roll dates) wrapped around a statistical core: the income term $C + \text{paydowns}$ is a prepayment-model output, so the implied financing rate inherits the model's error, and two dealers with different prepayment models compute different IFRs from the same two prices.

16.2 An illustrative computation

Suppose the drop is 12/32 (0.375 points) over 30 days on a price of 100, and coupon plus paydown income over the month is 0.45 points. The net financing cost is $0.375 - 0.45 = -0.075$ points: the roll pays the holder, an implied financing rate of about -0.9 percent annualized, while agency repo is positive (calibrated example; (Modelled, C)). Negative implied financing is the normal state for a special coupon, not an anomaly: it is the front contract's scarcity rent quoted as a rate.

16.3 Data status: not measurable here

The computation needs agency TBA price pairs by coupon and settlement month. Our mortgage holdings are the non-agency ABS-EE universe (registered CMBS and other structured products); the FINRA agency MBS sets we hold are trading-activity aggregates, not roll pairs. The relationship is therefore documented, with its measurement design specified, and not computed. We state the adjacent fact our data do support, because it is the roll market's settlement footprint: the EPN share

of MBSD clearing par (Section 6) measures how much agency settlement still moves specific pools, and its spring-2020 spike is the kind of observation a roll-specialness series would be validated against.

16.4 Verdict

Genuine, class 2 with model risk in its core quantity. Spurious only when the implied financing rate is quoted to false precision: the third significant figure of an IFR is a prepayment assumption, not a market price.

17 Perpetual futures funding

17.1 The relationship

A perpetual future has no expiry; the anchor between the futures price and the spot index is a periodic funding payment from the side that is crowded to the other, proportional to the premium. The mechanism is a proportional controller: funding $= \kappa \times$ (premium), paid every eight hours on the major venues, with the venue's funding cap bounding $\kappa \times$ premium at a maximum rate. Ackerer et al. (2024) derive the no-arbitrage price under general funding specifications (the futures price is the risk-neutral expectation of spot sampled at a random time whose intensity is the anchoring strength), and Angeris et al. (2022) give the design primer. The control-theoretic reading is the enforcement analysis: a proportional controller leaves a steady-state error that grows with persistent one-way flow and falls with the gain κ; and when the premium is large enough that the cap binds, the controller saturates and the anchor is, for that period, switched off. The relationship is class 2 (a contractual rule) whose strength is a parameter of the rule and of the venue's solvency.

17.2 Calibrated magnitudes

Neutral funding of 0.01 percent per eight hours is 10.95 percent annualized; sustained bull-market funding of 0.05 percent per eight hours is about 54 percent annualized. A $20 million delta-neutral book (long spot, short perpetual) at 0.05 percent collects $30,000 per day gross (Modelled, C). The gross figure is the marketing number; the net figure subtracts spot borrowing or opportunity cost, venue fees, and the basis risk between the venue's index and the hedge's execution venue, and it carries the loss states below at full size.

17.3 Breakdown

The breakdowns are institutional, not statistical: venue failure (the FTX collapse stranded the hedge leg and the collateral simultaneously), stablecoin depeg (the numéraire of the contract moves), and cross-venue index divergence (the anchor holds to the venue's index, which is not the market's price). In each case the funding rule performed as written and the relationship failed anyway, because the enforcer was the venue.

17.4 Data status

We hold no crypto market data. The library's DeFiLlama coverage (data_through 2026-06-03) is protocol-level aggregates, not venue funding rates. The section is framework-only, included because

the controller analysis is the cleanest statement in the catalogue of a general point: an enforcement rule with a cap has a known maximum strength, and deviations beyond the cap are unbounded by the mechanism.

17.5 Verdict

Genuine as a mechanism design; the funding stream is a real, contractual cash flow. Spurious as a yield quote: an annualized funding rate presented without the venue, depeg and saturation states is a number with its loss distribution removed.

18 Segmentation: onshore-offshore and dual-listed prices

18.1 The relationship

Where law reserves a market to a class of participants, the same claim trades at two prices, and the gap $|p_{on} - p_{off}| \leq \kappa$ is bounded only by κ, the shadow cost of crossing the boundary. Unlike every other band in this catalogue, κ is not a cost stack that arbitrageurs pay down; it is a policy price, set by the authority that maintains the boundary, and it can be moved to any value, including infinity, by administrative action. The gap is therefore a threshold process, not a mean-reverting one: it sits at whatever level the current enforcement intensity implies, and it closes when policy closes it, usually by making one leg unfundable rather than by letting arbitrageurs trade.

The documented episodes are consistent with that reading. Offshore renminbi (CNH) traded at discounts of 1.5 to 2 percent to onshore CNY for extended periods in August 2015, against a 2 percent onshore daily band; in January 2016 overnight CNH HIBOR printed above 60 percent, as reported at the time, which is the signature of the offshore leg being made unfundable to close the gap. The A-H premium (mainland versus Hong Kong listings of the same companies) has persisted for roughly two decades through every change in connect-scheme plumbing. Nigeria's multiple exchange-rate windows, the extreme case, were unified by decree in 2023: the segmentation ended by statute, not by convergence.

18.2 Data position

Our data hold the offshore leg only: the DTCC FX partition carries forwards and non-deliverable forwards by pair (the largest September 2026 NDF populations are KRW/USD with 262,819 trades, TWD/USD with 222,360, and INR/USD with 210,541 (Observed, A)), which measure offshore pricing and activity but contain no onshore price to differ against. The BIS effective exchange rate indices (USD nominal effective 117.47 in April 2026, 2021 = 100; real 117.70; via the Econ Data Library, data_through 2026-04-01) are context for the dollar's level, not observations of any segmentation gap. The relationship is documented with its measurement requirement stated (a paired onshore series under the same timestamp convention), and no gap is computed.

18.3 Verdict

Genuine as a policy object; spurious as a relative-value trade. A segmentation gap is a quoted price of a law. It has no half-life, its convergence is a political event, and backtests of it measure the stability of a policy regime while describing themselves as measuring a mispricing.

19 The municipal ratio

19.1 The relationship

Municipal interest is exempt from federal income tax, so the taxable-equivalent comparison defines an implied marginal tax rate τ^*: $y_{muni} = (1 - \tau^*) \cdot y_{taxable} - \text{credit} - \text{liquidity}$. The municipal ratio ($y_{muni}/y_{UST}$) is the market's running estimate of τ^* net of the other terms. The relationship is held in place by a clientele equilibrium, not by arbitrage: the marginal holder is the investor whose tax bracket clears the market, and τ^* moves with the bracket structure, with the supply of tax-exempt paper, and with the composition of demand (households directly, funds, banks and insurers under their own tax treatments). Wang et al. (2008) decompose the yield and estimate the liquidity premium at 9 to 13 percent of yield for AAA bonds and up to 19 percent for BBB, so on a 4 percent BBB yield about 0.76 percentage points is liquidity, not tax. Longstaff (2011), using tax-exempt swap rates, identifies an average marginal tax rate of 41.6 percent in his sample and a negative risk premium on the tax factor, consistent with after-tax cash flows being worth more in bad states.

The structural lesson generalizes beyond municipals, and it is why a relationship we cannot measure earns a full section: every spread in this catalogue is a sum of terms quoted as one number, and the municipal ratio is the case where the decomposition is forced on the analyst by the tax code itself. Quoting the ratio as a tax rate, without stripping credit and liquidity, is the same error as quoting a CDS-bond basis as mispricing without stripping funding.

19.2 The legislated break

The Tax Cuts and Jobs Act of 2017 is the catalogue's cleanest legislated break: it cut the corporate rate (reducing bank and insurer demand for exemption), cut individual rates (reducing τ^* mechanically), and repealed advance refunding (changing supply). A backtest of the municipal ratio across 2017 estimates a law change as if it were a sequence of mispricings, and any convergence statistic computed across it is uninterpretable. The protocol of Section 5.5 (rule 4) exists for this case.

19.3 Data status

We hold no municipal panel: the print store covers a 710-bond corporate universe and munis are outside it. The relationship is documented, not measured, with the decomposition and the break recorded. DTCC's underwriting file does carry the municipal issuance leg (Section 22, Figure 12): municipal amounts run \$0.38 to \$0.67 trillion per year over 2011-2025 (OBSERVED, A), the supply side of the clientele equilibrium, which is the one leg of this relationship our data touch.

19.4 Verdict

Genuine as a clientele equilibrium with a statutory core. Spurious as a convergence trade, and doubly spurious across 2017.

20 The loan-tranche-ratings triangle

20.1 Three restrictions on one dataset

A securitization maps a loan tape into a tranche stack by a waterfall. For tranche k with attachment a_k and detachment d_k, the loss borne is $\text{loss}_k = \min(\max(L_d - a_k, 0), d_k - a_k)$ where L_d is the pool loss of deal d: credit support at every point of the stack is a deterministic function of the loan tape and the deal's current balances. Ratings map collateral states into grades. The triangle is the joint system of three restrictions on the same data: (i) the waterfall holds exactly, so tranche balances and support computed from filings must reconcile with what trustees report; (ii) the collateral-state-to-rating mapping is stable within an agency and a vintage, so a fundamentals model estimated on one era should price the ratings of another; (iii) rating changes lag collateral changes, with a lag distribution that Rule 17g-7 disclosure makes observable. Restriction (i) is class 1 (identity). Restrictions (ii) and (iii) are class 5, and they are where the information is: the gap between what the collateral implies and what the agencies assign, and the delay with which they respond, are the raw material of surveillance. This is the only relationship in the catalogue where all three legs are observable at scale in our data, and it ranks first in Appendix B for that reason.

The universe is registered CMBS: 517 deals, 28,638 loans ($518.77 billion original balance), 11,103 tranches (Observed, A). Mean deal size is $1.00 billion (median $0.91 billion) with 55 loans per deal on average; the vintage mix is 226 pre-2020 deals, 123 from 2020-2022 and 168 from 2023 onward (Observed, A). The tranche CUSIP, the join key across the three legs, is populated for 99.06 percent of tranches (10,999 of 11,103); 76.4 percent of CUSIP classes (8,401 of 11,001) join to at least one agency's current rating (Observed, A).

20.2 Leg 1: the waterfall as an audit

The 10-D build parsed 23,464 of 23,636 inventoried distribution reports (172 extract-empty, no parser failures), giving 484 distribution dates from November 2012 to September 2026 and a balance history of 738,118 tranche-months, 96.2 percent spine-joined to the deal registry (Observed, A). Derived credit support (computed from balances, applying the waterfall) is populated for 94.3 percent of tranche-months in the current-support table (660,940 of 701,016 rows) (Observed, A). The audit content is in the discrepancies between derived support and trustee-stated support. Three anchors, verified against the statements:

- BANK 2019-BNK20, class A-2 (CUSIP 06540AAC5): original balance $350 million, current $283,310,699.19, factor 0.8095, trustee-stated support 33.16 percent (Observed, A).
- BANK 2021-BNK35, class A-2 (CUSIP 06540CAX5): original $90.7 million, current $28,499,999.89, factor 0.3142, stated support 32.09 percent (statement of 17 September 2026) (Observed, A).
- MSBAM 2017-C34, class A-3 (CUSIP 61767EAD6): current $233,714,159.10 on $250 million original, factor 0.9349, stated support 34.92 percent, derived support 72.69 percent (Observed, A).

The third anchor is the instructive one: a 37.8-point gap between stated and derived support on the same class at the same date. The difference is definitional, not fraudulent: the derived figure counts the balances of classes junior in the stack under one treatment, the trustee's stated figure

Table 8: Rating gap by agency: assigned notch less model-implied notch, senior class

Agency	Deals	Mean gap	Median gap	SD
Fitch	467	0.33	0.45	0.93
KBRA	348	0.15	0.29	0.92
S&P	194	0.20	0.24	0.51
Moody's	282	0.21	0.25	0.39
DBRS	121	0.21	0.44	1.14
Pooled	475	0.31	0.46	0.94

Notes: Positive gap: the assigned rating is weaker (stricter) than the fundamentals model implies. The pooled row averages agency gaps per deal over agencies rating that deal. Classification of the pooled gap: 50 deals rated above the model (more generous), 328 in line, 97 rated below the model (stricter), 42 not rated in the data. Source: ABS-EE and 10-D builds; 17g-7 event layer; fundamentals model as described in the text.

under another (the treatment of classes below the class in question, including exchangeable and strip classes). The triangle exists to reconcile exactly this: two support numbers for one tranche, each internally consistent, differing by the definitional treatment of the stack below. Across the build, 39 such definitional cells are carried as flags on as-filed values rather than resolved by overwriting either source. An identity that reconciles 94.3 percent of a 701,016-row table and isolates 39 definitional cells is doing audit work that no price-based method can do.

20.3 Legs 2 and 3: the rating gap and the response lag

For the second leg we estimate a fundamentals model in the style of Griffin and Tang (2012): loan-level characteristics (debt service coverage, loan-to-value, property type, occupancy, loan age and structure) feed a hazard model of severe delinquency and loss, pool expected loss is aggregated, and each deal's senior class is assigned the notch whose expected-loss band its fundamentals imply. The gap is assigned notch less model-implied notch on a scale where higher is weaker, so a positive gap means the agency is stricter than the fundamentals model. Table 8 and Figure 11 report the distribution across 517 deals (Modelled, B).

The pooled mean gap is +0.31 notches (median +0.46): post-crisis agencies rate registered CMBS senior classes about a third of a notch stricter than a fundamentals model estimated on the same collateral. Fitch is the strictest on average (+0.33) and KBRA the closest to the model (+0.15). The distribution is the finding, not the mean: 328 of 475 rated deals sit in line with the model, 97 are stricter, 50 more generous, and the left tail (Figure 11) contains the deals where model and agency disagree by two to five notches, which are the surveillance queue. Senior attachment averages 28.2 percent of the pool, and modeled subordination exceeds model-required subordination by 0.375 on average in the build's units (Modelled, B): the post-crisis stack carries more protection than the fundamentals model requires, which is the balance-sheet counterpart of the rating gap's sign.

For the third leg, timing: in the build's first-downgrade race at a three-year horizon, agencies reach downgrades before the fundamentals model does (Fitch AUC 0.914 against the model's 0.731 for first-downgrade prediction), while the fundamentals model alone attains 0.692 AUC for severe outcomes at three years (deal-level specification 0.723; the loan-level three-cause extension 0.701 at three years and 0.610 at one year) (Modelled, B). Agencies are not asleep; they are faster on their own actions than a collateral-only model, which is expected, since they see servicer information the tape does not carry. The surveillance content of the triangle is therefore not that agencies are slow in general, but the

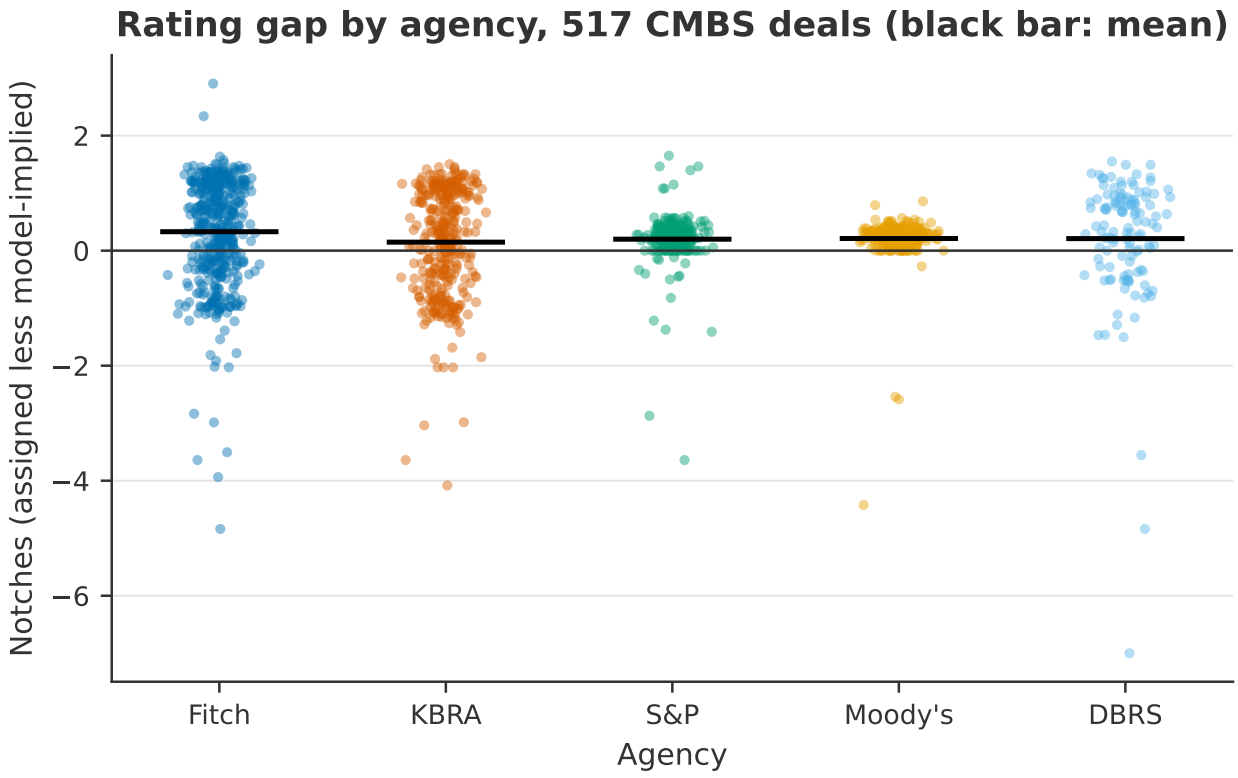


Figure 11: The rating gap by agency across 517 CMBS deals

Source: ABS-EE, Form 10-D and Rule 17g-7 builds; authors' fundamentals model. Note: Each point is one deal's senior-class gap for that agency; the black bar is the agency mean. Most mass lies within one notch of the model, with a long left tail of deals the model scores far stronger than the agencies do.

interaction of legs 2 and 3: deals where the collateral has moved, the gap has widened, and no action has followed. That interaction is only measurable while actions are observable, and they stop being observable soon: every large agency's Rule 17g-7 file ends in October or November 2025, while the collateral tape runs to October 2026. There is roughly a year of collateral history with no observable rating response, and the gap accumulated in that window is the triangle's current output: a queue of deals ranked by collateral deterioration, awaiting either agency action or a substitute disclosure.

20.4 Breakdown

The triangle breaks where its inputs do. Its universe is registered offerings only: the large 144A issuers (including the datacenter ABS complex) file only Form ABS-15G and are outside ABS-EE entirely, so no triangle statement extends to them. Appraisal reductions and servicer watchlist status live in the 10-D statements at statement level (an anchor case: JPMDB 2020-COR7 carries an appraisal reduction of $22,625,176.00 and cumulative ASER of $639,972.89, visible in its statement but not in the loan tape), so the collateral leg understates servicer knowledge by construction. And restriction (ii) fails by design in one direction: agencies rate through the cycle with qualitative overlays, so a stable mapping is a null to be measured against, not a truth.

20.5 Verdict

Genuine, and the catalogue's most productive relationship: an identity that audits, a gap that ranks, and a lag that times. Its claims are the most falsifiable in the paper, and Appendix B's Test 1 was built to falsify the central one (that the collateral block adds out-of-sample predictive content for rating actions beyond the current rating) on a 2023-2025 holdout. The test has been run: the collateral block added 0.0092 AUC, below the pre-specified 0.01 threshold, and the claim is withdrawn (Appendix B).

21 Holder-implied pricing: fund marks against prints

21.1 The relationship

For illiquid bonds, two independent measurement systems produce numbers for the same object. The transaction system produces prints $p_b(t)$, the last TRACE trade price. The accounting system produces marks $m_b(t)$, the fair values that funds report on Form N-PORT, of which we take the cross-fund median with its own report period and holder count. Define two gaps. The mark-print gap $g_1 = m_b(t) - p_b(t)$ in price points, and the derivative gap g_2, the difference between the price implied by the issuer's CDS curve and the print. The g_2 leg awaits the Section 12 join and is not computed here; this section is about g_1, and about the fact that g_1 supports two opposite readings. Under the *appraisal* hypothesis, marks are smoothed appraisals: mark changes have lower variance than print changes, marks lag prints, and g_1 is positively autocorrelated measurement error that decays as appraisals catch up. Under the *signal* hypothesis, holders know something: g_1 predicts the direction of future prints, because informed holders' accountants mark closer to the value the market will print next. The hypotheses have opposite trading implications (fade the mark versus follow it) and are separable in the data, which is what makes the relationship a research object rather than a curiosity.

The measurement base is joint: 550 of the 707 monitored bonds (78 percent) carry a fund mark in the latest report period and 603 in at least one quarter, against a print store of 4,100,179 trades (Observed, A). Event-time discipline is the first design requirement (Section 5.3): a mark is stamped at its fund report date, and any test must align the print clock to it. Our own pipeline supplied the cautionary example: an earlier construction stamped marks with the SEC report-ending-period field, which placed 935 bond marks at future dates; the correction (marks restamped to report date; 5,699 labels changed; values and the derived already-below set unchanged) is documented in the data appendix because it is exactly the error this section's tests are designed around.

21.2 Measured instances, and the confound that governs them

The largest gaps in the current panel are in convertible and equity-linked structures, and they are large: TeraWulf CUSIP 88080TAB0 printed 171.7518 against a median mark of 305.7621 for the period ending 30 June 2026, a gap of 134.01 points; its sibling 88080TAC8 printed 125.1394 against 213.8395 (gap 88.70); IREN CUSIP 46270CAB5 printed 184.0207 against 291.239 (gap 107.22) (Observed, B). Prices are percent of par, and values far above 100 are coherent for these instruments. But that coherence is the confound: a convertible's mark embeds the conversion option's value at the marking model, while a print is a transaction in the whole instrument at a moment, and the two need not embed the same equity volatility, the same borrow, or the same credit. A naive ranking of $|g_1|$ therefore ranks embedded equity optionality and marking-model dispersion, not appraisal error. Stripping the option (valuing the conversion feature under a common model and netting it from

both sides) is the first task of any test of the appraisal-versus-signal question, and until it is done the large-gap tail of the panel is uninterpretable. We regard this as the section's main result: the relationship's measurement problem is identified, located in a specific instrument class, and removable, which is more than most unresolved relationships offer.

For straight senior bonds the gaps are smaller and the two hypotheses are cleanly testable on the panel as built: the mark series is quarterly, the print series daily, and the appraisal hypothesis makes a sharp variance prediction (mark changes smoother than print changes over matched windows) that the signal hypothesis denies.

21.3 Convergence, persistence, breakdown

Nothing enforces $g_1 \to 0$: marks and prints are answers to different questions (what would we carry it at; what did it trade at), produced under different rules (fair-value standards versus transaction reporting). The gap's persistence is the persistence of those rules and of the instruments' liquidity. It breaks at events that force the two systems to the same number: a large print after a long silence, a fund liquidation, an index inclusion, a default.

21.4 Verdict

Unresolved by design, and stated as such: the relationship is a well-posed question with a built panel, an identified confound, and a test (Appendix B does not carry it; it is the fourth-ranked item) that has not yet been run. We include it at full length because the catalogue's discipline requires the unresolved items to be as precisely specified as the resolved ones; a gap of 134 points that turns out to be an option is the same lesson as Section 13's 33 bp that turned out to be a basket.

22 Persistence, regime shifts and costs across the catalogue

22.1 Persistence is a property of the class

Table 9 collects the persistence character of the sixteen relationships. The pattern the framework predicts is the pattern the catalogue shows: identity ratios drift at the speed of institutional change (era means 5 to 10 points apart, Section 6); contract relationships converge on calendars (settlement, roll and funding dates) and hold their levels between them; the balance-sheet band relationships hold levels with small within-regime variance (the 10-year swap spread's monthly standard deviation is 2.3 bp) and jump when the band reprices; physical conversion spreads persist for the life of a capacity outage; statistical relationships have no convergence clock at all. A single half-life estimated across the catalogue would average objects that do not share a clock.

22.2 Regime shifts: the four kinds

The catalogue's histories contain four distinct kinds of break, and they must not be pooled. *Legislated* breaks change the enforcement itself: the fails charge (2009), the TCJA (2017), the Nigerian unification (2023), clearing mandates (which moved the netting ratio's level around 2011-2012 as newly cleared flow arrived). *Balance-sheet* breaks reprice the class-4 band without changing any rule: March 2020 is the template, and the spring 2026 fails spike sits in the same family at smaller scale. *Supply* breaks move class-5 factors: the 2026 Treasury repricing of Section 9 (real yields +33 bp in seven weeks on

Table 9: Persistence and cost character by relationship

Relationship	Clock	Within-regime behaviour	Dominant cost
Netting identity	Institutional (years)	Era-level drift	None (audit object)
Swap spread	Funding / issuance (quarters)	Level, SD about 2 bp/mo	Balance sheet, margin
CIP basis	Reporting calendar	Level with calendar spikes	Balance sheet
Convenience	Supply (years)	Slow drift, crisis spikes	Not isolable
Specialness	Auction cycle (weeks)	Issue-level decay	Fails charge cap
Cash-futures basis	Delivery date	Converges if funded	Repo funding
CDS-bond basis	Funding (quarters)	Band, widens in stress	Funding both legs
Index additivity	Continuous	Small on correct population	Replication funding
Correlation premium	None	Premium, crash state	Margin in the crash
Conversion spreads	Capacity (years)	Rent while capacity out	Physical carry
Dollar roll	Roll calendar (monthly)	Specialness by coupon	Model (prepayment)
Perp funding	Funding interval (8h)	Collects until saturation	Venue and basis risk
Segmentation	Political	Level until decree	Boundary is absolute
Municipal ratio	Tax years / statute	Clientele level	Liquidity term
Ratings triangle	Servicer / agency (months)	Gap accumulates, actions lag	None (surveillance)
Holder marks	Reporting (quarterly)	Smoothed, event breaks	Option strip required

Notes: Authors' classification from Sections 6 to 21.

flat breakevens) is a supply repricing propagating through convenience into every convenience-sensitive residual. *Measurement* breaks change the record, not the market: the FICC file's unit convention change, its mid-2011 settled-level shift, and our own N-PORT mark restamping are breaks in e_t in the notation of Section 5.2, and a regime test that cannot tell a measurement break from a market break will find regimes in the data pipeline. The underwriting record (Figure 12) illustrates a series that must be read with all four in mind: recorded amounts step up in 2024-2025 (total $4.86 trillion in 2024 and $5.94 trillion in 2025 against $3.62 trillion in 2023 (OBSERVED, A)), driven by corporate counts roughly tripling (10,585 in 2022 to 35,877 in 2025), a pattern consistent with coverage or classification change in the source at least as much as with issuance behaviour; we report the series as recorded and do not build on the step.

22.3 Costs: the three layers

Every tradable relationship in the catalogue pays three layers of cost, and the sections' verdicts can be restated as statements about which layer binds. The *spread* layer (bid-offer on both legs) binds for the liquid relationships (index additivity, dispersion) and is the only layer most backtests model. The *funding* layer (repo, margin, leverage-ratio rent) binds for the class-4 relationships and is the equilibrium itself there, not a friction on top of it: Section 7's arithmetic has the funding layer at roughly twice the carry at the measured spread. The *event* layer (the loss state: the basis unwind, the correlation spike, the venue failure, the decree) binds for the class-5 and funded relationships and is systematically absent from backtests, because the event is rare in-sample by construction. A catalogue-level rule follows: quote every relationship's expected return net of the layer that binds for its class, and treat any result quoted net of only the spread layer as an upper bound labelled as such.

22.4 What the catalogue says jointly

Read together, the sixteen sections make one joint claim that none makes alone: in the autumn of 2026, the measurable relative-value surface of US markets is dominated by the price of balance sheet and the price of collateral scarcity, both of which are institutional parameters before they are market

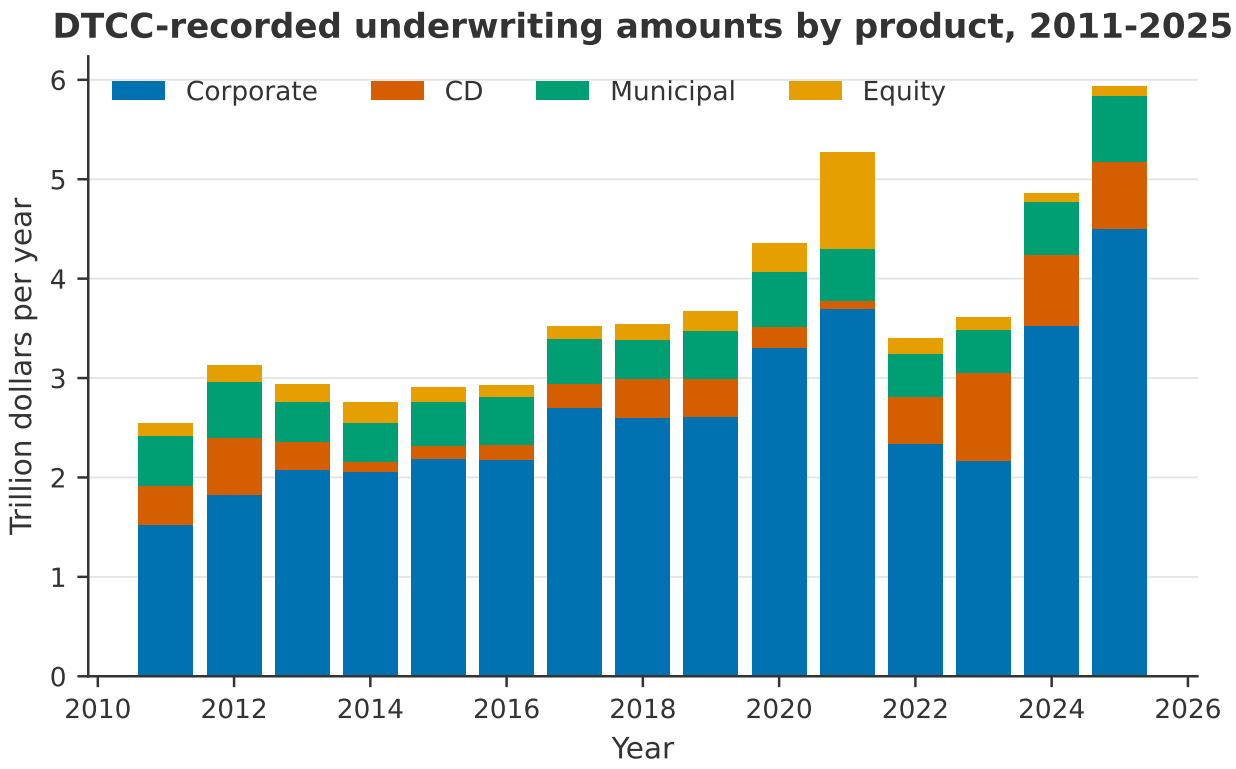


Figure 12: DTCC-recorded underwriting amounts by product, 2011-2025

Source: DTCC underwriting file (DTCC, 2026a). Note: Amounts as recorded by the source. 2026 (January to August, $4.58 trillion) is excluded as a partial year. The 2024-2025 step coincides with a near-tripling of recorded corporate counts and should be read as recorded coverage, not necessarily as issuance.

prices. The swap spread (−39 bp at ten years), the CIP basis (literature range −10 to −41 bp), the convenience decomposition (a 33 bp real repricing on supply), the GCF gap (−0.4 bp, no aggregate scarcity), fails at 0.012 percent of cleared flow, and a netting multiplier of 3.06 are one observation from six angles: intermediation capacity is the scarce factor, it is priced, and its price is stable within the regime. The relationships that would contradict the claim (a wide GCF gap, fails at cascade levels, a positive swap spread) are absent from the data. That is the paper's empirical bottom line.

23 Limitations and robustness

Falsifiability status. This paper is a retrospective catalogue and measurement exercise. No analysis plan was registered before the data were examined, and the descriptive statistics in Sections 6 to 21 should be read as measurements, not as tests. The exceptions are the three designs in Appendix B, which were stated in full, with falsifiers and holdout samples, before those holdouts were touched. Test 1 has been run and is reported in Appendix B against its stated thresholds: the collateral block added 0.0092 AUC on the 2023-2025 holdout, below the pre-specified 0.01 threshold, and the claim is withdrawn. Tests 2 and 3 have also been taken to their stated endpoints and are reported in Appendix B against the stated thresholds: Test 3 was run, two of its falsifier clauses triggered, and its claim is withdrawn; Test 2 could not be executed as designed, because the fails data its falsifier requires do not exist for the 2020-2026 evaluation window, and its claim is therefore neither supported nor withdrawn.

The catalogue's selection (sixteen featured relationships from a larger examined set, Appendix A) is itself a multiplicity problem, handled by the protocol of Section 5.5 rather than eliminated.

Depth of the tape. The swap tape measurements rest on one complete full-fidelity month (September 2026), with July, August and partial October slices for the credit panel. Levels measured in one month cannot distinguish a regime from a month; we mitigate by anchoring the swap-spread reading in a break-even calculation whose inputs are institutional (margin schedules, haircuts, hurdles) rather than estimated from the same month, and by labelling every tape statistic with its window. The parse extends backward month by month; the September levels will be re-based when a second full quarter exists, and the convenience factor test in Section 9 requires exactly that.

Tape construction. Three tape features bound what can be claimed. Prices are reconstructed from class-specific fields because the dissemination price field is unpopulated; a field-population change by the repository would alter reconstructions without altering markets, and our field census (Section 4) is the control. Notionals are capped on two-thirds of SEC credit trades, so every notional-weighted statistic is computed on floors and is reported as such; medians of spreads, our main objects, are unaffected by the caps. The equity partition's underlier identity is unpopulated, which blocks Section 14 entirely and any equity-linked decomposition elsewhere.

Joins not yet built. The CDS-bond basis (Section 12) and the index additivity computation (Section 13) both await joins (CDS names to bond prints; index constituents to the name panel). We have reported the legs and the naive exhibits rather than build the joins hastily inside this paper, because both joins involve name-resolution judgement whose errors would enter the results silently. The CoreWeave boundary case (widest monitored credit story, absent from the CDS population) is a reminder that join coverage, not join precision, is the binding constraint.

Data that are not public at trade level. Index tranche quotes, issue-level specific-collateral repo rates, TBA roll pairs, listed option prices and perpetual funding rates are not in the public record we can assemble. Sections 13 (tranche leg), 10 (issue level), 16, 14 and 17 are correspondingly partial. We have not substituted proxies for any of them; the data appendix records each absence as an absence.

Series breaks in sources. The FICC volumes file changes unit convention across its history (handled by magnitude normalization and anchor-verified) and its settled series shifts level around mid-2011 for reasons the file does not document; era comparisons across that point are provisional. The underwriting file's 2024-2025 step is reported as recorded. The GCF and fails files are rolling 250-day windows whose history before October 2025 does not exist in the source's current publication, so no statement in Section 10 extends beyond that window except through the monthly FICC file. The ratings leg's observability ends in October and November 2025 by the agencies' filing behaviour, not by our choice; the triangle's third leg is unmeasurable after that date until filings resume or an alternative disclosure is built.

Model dependence in the triangle. The rating gap of Section 20 is measured against a fundamentals model; a different defensible model moves the gap. We mitigate in three ways: the gap's sign and ranking are stable across the build's specifications (the pooled mean is +0.31 notches in the reported specification, and the agency ordering is preserved in the alternatives the build records); the waterfall leg is model-free, so the audit results stand independently of the gap; and Test 1 in Appendix B is designed so that the model's usefulness, not its truth, is the estimand (out-of-sample predictive content for agency actions).

Robustness of the headline numbers. The netting anchors (January 2001 and September 2026 ratios) are recomputed from raw values in the source file and are exact. The swap spreads are medians of daily trade medians; using trade-weighted means instead moves the 10-year figure by under 2 bp in September, inside the band the break-even calculation spans. The convenience decomposition is two

FRED series differenced; its robustness is the flatness of the breakeven (0 to +2 bp depending on the end date within 20-21 October), which does not touch the conclusion. The naive index gap is stable in the worst way: it does not move under any variation we tried, because it is composition.

24 Conclusion

We set out to organize relative value by enforcement rather than by asset class, and to measure what a public-data record can measure under that organization. The organization held up: the sixteen relationships sort into five classes whose members share dynamics, failure modes and cost structures, and the sort, once made, does most of the analytical work. Identities audit. Contracts converge on calendars and break by legislation. Conversion spreads pay capacity rents. Banded no-arbitrage relationships quote the price of balance sheet, hold their levels, and widen together when that price rises. Statistical relationships pay premia for named loss states and have no convergence clock.

The measurements support three conclusions. Netting, the plumbing identity under the cleared system, is a slow state variable whose current configuration (GSD multiplier 3.06, EPN settlement share 2.6 percent) differs materially from its configuration in 2001 (3.37 and 19.2 percent), and any risk arithmetic that carries it as a constant is wrong by a first-order amount. The balance-sheet band is measurable on the public tape, and in September and October 2026 it sits where the costing says it should: swap spreads of -13 to -70 bp by tenor against a break-even near -30 bp at ten years, a real-yield repricing of 33 bp on unchanged inflation expectations, no aggregate collateral scarcity in the GCF complex, and fails at a hundredth of a percent of cleared flow. And the largest apparent opportunities in the data are artefacts: a -33 bp index gap that is a basket definition, and 134-point mark gaps that are an embedded option. The discipline of naming the enforcer and the population before computing anything is what separates those from the relationships worth capital.

The three tests of Appendix B have been resolved on their stated terms. The first test's falsifier fired: the triangle's collateral block added 0.0092 AUC on the 2023-2025 holdout, below its pre-specified threshold, and the claim is withdrawn. The third test, the balance-sheet factor across the swap-spread and CIP panels, was run, and its claim is also withdrawn: two falsifier clauses triggered, on calendar coefficients that are insignificant with the year-end coefficient wrong-signed, and on a cleared-versus-bilateral contrast that is absent. The second test, the netting ratio's predictive content for settlement stress, could not be executed as designed: the fails data its falsifier requires do not exist for the evaluation window, and the claim stands neither supported nor withdrawn. The forward agenda for that test is time rather than analysis: the standing daily collection of the fails file is accumulating the history the design needs, and an honest evaluation opens in 2028. The falsifiers are the point: a catalogue of relationships that cannot be wrong about anything is a list of stories. Two data acquisitions would change the catalogue's coverage more than any further analysis of what we hold: index constituent membership joined to the tape (which converts Section 13's exhibit into a measurement and Section 12's basis into a number), and the resumption or replacement of Rule 17g-7 disclosure, without which the ratings leg of the triangle is running on a clock that stopped in the autumn of 2025.

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A Examined and rejected

The catalogue's sixteen relationships were selected from a larger examined set. The rejections are recorded here with their reasons, because the selection is part of the method (Section 5.5) and because several rejections are close calls that a future dataset could reverse.

Pairwise cointegration screening over the CDS name population. Rejected on arithmetic. Screening 1,251 names pairwise is about 782,000 pairs; at a 5 percent level the null alone manufactures roughly 39,000 false positives, and the panel offers only four monthly slices, so no pair has the time-series depth a cointegration claim needs. The screen would return a ranked list of artefacts with standard errors. Revisit only with a multi-year daily panel and a pre-registered multiple-testing correction, at which point it becomes a different study.

Equity index lead-lag and opening-gap effects. The effects are real in the literature and dead after costs at our data's resolution: we hold no intraday equity prices, and the daily record cannot express the trade. Rejected for absence of the required data, not for absence of the phenomenon.

Variance risk premium harvesting as a featured trade. The premium is genuine (Section 14's cousin), but as a featured relationship it fails screen 2 in its usual presentation: the loss state is the portfolio's worst state, and without options data we cannot size the position or the premium. Retained only as the correlation premium's framework discussion.

Carry as a universal cross-asset factor. Rejected as a category error under the framework: pooling carry across enforcement classes (funding carry in class 4, term carry in class 5, conversion carry in class 3) produces a factor whose composition changes with the sample, so its estimated premium is a mixture weight, not a price. Each class's carry is treated inside its own section.

Transfer entropy for lead-lag. Rejected on measurement: estimates are dominated by discretization and binning choices at our sample sizes, and we know of no documented case where transfer entropy beats linear Granger causality at equal data on financial series of this length. The identified lead-lag questions in the catalogue (ratings lag, mark lag) are institutional and are estimated as lag distributions instead.

Machine-learned spread prediction without a mechanism screen. Rejected by the catalogue's founding rule. A flexible learner on our panels will find reporting dates, roll dates, tax years and file-format changes, and will present them as alpha with excellent in-sample fit. Section 13's naive gap is the deterministic version of the same failure. Learning methods are admissible downstream of the mechanism screen (the hazard model of Section 20 is one), not upstream of it.

Textbook Merton capital-structure arbitrage (equity versus CDS on the same issuer). Rejected on the evidence of its canonical episode: the May 2005 GM trade, in which the apparent mispricing was mostly recovery and seniority assumption risk expressing itself exactly when the rating event arrived. The gap between a structural model's implied spread and the market spread is dominated by the model's recovery and capital-structure inputs, which are the unobservables the trade claims to arbitrage. The relationship fails screen 1 in practice: the enforcer would have to be a trader who knows recovery better than the market, which is an information claim, not an enforcement mechanism.

B First tests for the three ranked relationships

The three designs below were stated in advance of their holdout samples being touched, with falsifiers and thresholds fixed here. They are the paper's only pre-specified tests (Section 23). All three have

since been taken to their stated endpoints, and each outcome is reported with its design below: Test 1 was run and its claim withdrawn; Test 2 proved not executable as designed on the data that exist, and its claim is neither supported nor withdrawn; Test 3 was run and its claim withdrawn.

Test 1: the triangle (Section 20)

Question. Does the collateral state of a CMBS tranche predict rating actions beyond what the tranche's current rating already contains?

Design. Discrete-time hazard model on the tranche-month panel (738,118 tranche-months, 2012-2026), with agency and vintage fixed effects and the current rating carried as state. Two specifications: ratings-only (current rating, age, agency, vintage) and ratings-plus-collateral (adding current derived support, support change over six months, deal delinquency state, DSCR and LTV state variables from the loan tape). Outcome: downgrade of at least one notch within the following twelve months. Estimation on 2012-2022; holdout 2023-2025, untouched until the specification is frozen. Metrics: AUC and Brier score on the holdout.

Falsifier. The claim is withdrawn if the collateral block adds less than 0.01 AUC out of sample, or if the collateral coefficients are jointly insignificant at the 1 percent level after Bonferroni adjustment across the block. A negative result is a publishable output of this design: it would establish that agency ratings already impound the public collateral tape, which is itself a finding about disclosure efficiency.

Outcome. Test 1 was run on 24 October 2026. The unit of observation is the tranche-agency-month, the unit the design's agency fixed effects and per-agency rating state require. The estimation sample is 193,450 tranche-agency-months with 3,031 downgrade events. One implementation restriction is material: the loan tape, and therefore the collateral covariates, begins in November 2016, so the effective estimation window is November 2016 to 2022 rather than the 2012 start the design anticipated; the restriction binds both specifications identically because both are estimated on the same complete-case sample. The primary holdout, after the protocol's full-window observability rule and the exclusion of 2023 to 2025 vintage deals whose vintage effects were never estimated, is 146,438 tranche-agency-months with 10,080 downgrade events (base rate 0.0688). On that holdout, the ratings-only specification attains AUC 0.8763 and Brier score 0.0560, and the ratings-plus-collateral specification attains AUC 0.8855 and Brier score 0.0542. The collateral block therefore adds 0.0092 AUC out of sample, below the pre-specified 0.01 threshold, so the AUC leg of the falsifier fails. The significance leg passes: the joint Wald test of the collateral block gives chi-squared (5 degrees of freedom) 926.65 with p equal to 4.535e-198, against a Bonferroni level across the block of 0.0020 and a Bonferroni-adjusted minimum individual p of 5.416e-175. Because the falsifier is disjunctive, failure of either leg withdraws the claim, and **the claim is withdrawn**. The plain reading is the one the design named in advance: the collateral block is strongly associated with rating actions in sample and adds a small amount of out-of-sample discrimination, but not enough to clear the stated threshold, which is consistent with agency ratings already impounding nearly all of the public collateral tape's signal.

Test 2: the netting ratio as a state variable (Section 6)

Question. Does the ratio component of netting, as distinct from volume, predict settlement stress?

Design. Local projections at horizons of one to six months, 2020-2026 evaluation window: monthly fails totals (aggregated from the daily file, and from the monthly FICC record for the pre-window period) and the GCF Treasury less MBS gap, each projected on lagged changes in ν, with ν decomposed

into a volume component (the projection of ν on compared volume) and a ratio component (the residual). Controls: compared volume growth, issuance calendar, quarter-end dummies. Placebo: gross volume alone, which must not predict if the ratio interpretation is correct.

Falsifier. The claim is withdrawn if the ratio component carries no out-of-sample predictive content for fails at any horizon in the window (rolling-origin evaluation), or if the volume placebo predicts as well as the ratio component.

Outcome. Test 2 was assessed for executability before any projection was estimated, and **the test is not executable as designed**. The regressor side of the design is complete: the netting ratio ν is present for all 81 months of the 2020-2026 evaluation window, and for the full span from January 2001 to September 2026 (309 months). The fails leg does not exist for the window. The DTCC daily fails file is a rolling window of 250 trading days (20 October 2025 to 20 October 2026); it yields 12 monthly fails totals inside the window, 11 of them complete, and no fails observation at all before October 2025, so no rolling origin can be formed for the first five years of the window and no out-of-sample evaluation is possible at horizons of one to six months. The design's named source for the pre-window period, the monthly FICC record, carries compared and settled volumes and clearing par only: it contains no fails field in any period. No New York Fed or primary-dealer fails series exists anywhere in the estate. The GCF gap, the design's second target, cannot carry the verdict, because every verdict condition, including the falsifier, is stated on fails; its own coverage also has a nine-month hole (January to September 2025) between the DTCC GCF workbook and the daily file. No partial leg was run and presented as a result. The outcome is a finding about data availability, not about the claim: **the claim is neither supported nor withdrawn**. Two routes remain. The design can be run unchanged once the standing daily collection of the fails file has accumulated enough history: monthly fails exist from October 2025, and a rolling-origin evaluation with a minimal training window needs on the order of 24 months of fails before its first out-of-sample origin, which puts an honest evaluation opening in 2028. Alternatively, the New York Fed's weekly primary-dealer fails series (published from July 2013) could be substituted, but that series measures dealer-reported fails across Treasury, agency and MBS positions rather than GSD clearing fails, a different population and a different concept of settlement stress, so substituting it would be a new, amended test requiring re-specification, not an execution of this design.

Test 3: one balance-sheet factor (Sections 7 and 8)

Question. Are the swap spread and the cross-currency basis two prices of one factor?

Design. Build two monthly panels from the tape as the parse extends: tape swap spreads by tenor (2, 5, 10, 30 years) and cross-currency bases by currency for the G10 pairs, each split into cleared and bilateral subpanels. Standardize each series; extract the first principal component of the pooled panel. Calendar regression of the factor on quarter-end and year-end dummies with Treasury issuance controls.

Success conditions, all required. The first component carries at least 60 percent of common variance; loadings have the predicted signs (wider negative spreads and more negative bases load together); calendar coefficients are positive (widening) and larger at year-end than quarter-end; and the widening is larger in the bilateral subpanel than the cleared subpanel.

Falsifier. The unification claim is withdrawn if the first component carries under 40 percent of common variance, or if the calendar coefficients are insignificant or wrong-signed, or if the cleared-versus-bilateral contrast is absent. The bilateral-cleared contrast is the discriminating observation: demand-side accounts do not predict it.

Outcome. Test 3 was run on the longest panel the tape parse supports: 18 monthly observations, April 2025 to September 2026, containing exactly one December. The calendar identification is therefore thin, and the year-end coefficient rests on a single month; that limitation is stated here rather than softened, and the verdict follows the pre-registered conditions mechanically on this sample. The first principal component of the pooled panel carries 55.8 percent of common variance, below the 60 percent success bar and above the 40 percent falsifier line: on the variance clause alone the result lands in the design's indeterminate band. Two falsifier clauses trigger independently. First, the calendar coefficients on the pooled factor are insignificant, and the year-end coefficient is wrong-signed: quarter-end $+0.60$ ($p = 0.73$) and year-end -0.98 ($p = 0.77$), against success conditions requiring positive coefficients, larger at year-end than at quarter-end. Second, the cleared-versus-bilateral contrast is absent on the quarter-end leg and reversed in sign: quarter-end widening is -0.09 in the bilateral spread subpanel against $+0.32$ in the cleared subpanel. The sign condition also fails: the AUD (-0.69), NOK (-0.45) and SEK (-0.14) bases load against the factor, and the bilateral 5-year swap spread loads at only 0.10. None of the four success conditions is met, and **the claim is withdrawn**. One structural feature of the tape is disclosed rather than designed around: the cleared basis subpanel is empty in the data, because cross-currency basis trades are effectively all bilateral (39 cleared basis trades across the whole window, against roughly 207,000 bilateral), so the cleared-versus-bilateral contrast is evaluated on the swap-spread subpanels. What survives is narrower than the claim: a strong common factor in the cleared swap spreads and the major bases exists, with the four cleared spreads loading between 0.92 and 0.96 and the JPY, CHF and EUR bases loading with them. Its calendar behaviour, its subpanel contrast and its uniformity across all G10 bases are not supported.

C Data Appendix

All series were retrieved on 21-22 October 2026 unless otherwise noted. "Limitation" states the binding constraint on use, not a general caveat.

Series	Source	Coverage	Limitation
FICC clearing volumes (GSD compared/settled; MBSD clearing par/EPN current face)	DTCC, file served from cms-prod.dtcc.com (DTCC, 2026b)	Monthly, Jan 2001 to Sep 2026; 309 months per division	Unit convention changes across the file (decimal trillions; full dollars in 2025-2026), normalized by magnitude and anchor-verified. Settled series shifts level around mid-2011, cause undocumented in the file.
GCF repo index (Treasury and MBS par, weighted average rates)	DTCC daily file (DTCC, 2026a)	Daily, 16 Oct 2025 to 16 Oct 2026; 250 rows	Rolling 250-day window maintained by the source; no earlier history is published in this file. (A separate workbook of daily rates 2005-2024 is held in the vault; the two do not overlap.)

Series	Source	Coverage	Limitation
Treasury and agency fails	DTCC daily file (DTCC, 2026a)	Daily, 20 Oct 2025 to 20 Oct 2026; 250 rows	Rolling window, as above. Aggregate by class only; no issue-level fails.
Sponsored Service volumes	DTCC daily file (DTCC, 2026a)	Daily, 16 Aug 2021 to 15 Oct 2026; 1,289 rows	Earliest rows report zero volume. A companion monthly file (Jul 2019 to Mar 2020) leaves Apr 2020 to Jul 2021 uncovered by either file.
Underwriting counts and amounts (municipal, corporate, equity, CD)	DTCC underwriting file (DTCC, 2026a)	Monthly, Jan 2011 to Aug 2026; 188 entries	Amounts as recorded; the 2024-2025 step coincides with a near-tripling of recorded corporate counts and is treated as recorded coverage.
Swap dissemination tape	DTCC PPD (pddata.dtcc.com), CFTC and SEC partitions (DTCC, 2026c)	Sep 2026 in full (60,245,115 trades); credit slices Jul, Aug, Oct 2026 (Oct through 19 Oct)	Dissemination price field unpopulated (prices from class-specific fields). Notionals capped on 66.2 percent of SEC credit trades. Equity underlier identity unpopulated. Full-history parse in progress, newest first.
Single-name CDS panel and sector baskets	Tape credit partitions; authors' sector classification of 272 entities	Monthly slices as above; 127 September members	Membership requires at least three priced 5-year trades; membership churns month to month, so basket moves mix price and composition.
Swap rate grid (median fixed rates by tenor)	Tape rates partitions, statistics layer	Daily, Jun to Sep 2026	Trade medians at exact tenor buckets, not a fitted par curve; October grid not yet built at writing.
Treasury par yields; breakeven inflation; WTI	FRED: DGS2, DGS5, DGS10, DGS30, T10YIE, DCOILWTICO (Federal Reserve Bank of St. Louis, 2026)	Daily; quoted window 25 Aug to 21 Oct 2026	The ICE swap-rate series ICSAADR10Y was pulled, found inconsistent with market readings (implied spread near +270 bp), flagged as a definition mismatch, and not used.

Series	Source	Coverage	Limitation
ABS-EE loan tape	SEC EDGAR, Form ABS-EE (Reg AB II) (U.S. Securities and Exchange Commission, 2026)	Filings Nov 2016 to Oct 2026; 517 deals, 28,638 loans	Registered offerings only. 144A issuers are absent by construction. 34 filings permanently unparsed with EDGAR-side evidence.
Distribution reports	SEC EDGAR, Form 10-D (U.S. Securities and Exchange Commission, 2026)	23,636 inventoried, 23,464 parsed; statements 2012 to Sep 2026	172 filings extract-empty. Appraisal reductions and watchlist data present at statement level only. One trust (CIK 1541480) uncovered.
Rating events	SEC Rule 17g-7 agency files (U.S. Securities and Exchange Commission, 2026)	28,292,725 deduplicated events	Agency files end Oct-Nov 2025 at every large agency; rating actions after those dates are unobservable in this source.
Fund holdings and marks	SEC Form N-PORT	28 quarters; 707-bond monitored universe	Marks labelled by fund report date. An earlier labelling by report-ending-period placed some marks in the future; corrected 20 Oct 2026 (5,699 labels; values unchanged).
Bond prints	FINRA TRACE via the monitored collection	4,100,179 trades; 707 bonds (710 registry entries)	Monitored universe only (datacenter and adjacent issuers), not the corporate market. Append-only store.
Context macro series (euro-area AAA curve; BIS effective rates; EIA energy)	ElkassabgiData Econ Data Library, producer-first	ECB data_through 2026-06-23; BIS 2026-04-01; EIA 2026-05-26	Library lags primary publishers by weeks; used for context and discovery only, never as the sole source for a material figure.